

Linear differential Equations with Constant Coefficients (H.O)

① Complementary function

$$\frac{d^n y}{dx^n} + p_1 \frac{d^{n-1} y}{dx^{n-1}} + p_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + p_n y = X$$

Solve.

linear differential equation with constant coefficient.

$$\underline{\underline{\quad\quad\quad}} = 0$$

find CF

operator

$$\left(\frac{d}{dx} = D \right)$$

$$\frac{d}{dx} = D$$

$$\frac{d^2 y}{dx^2} = D^2 y$$

$$\frac{d y}{dx} = D y$$

$$\frac{d^3 y}{dx^3} = D^3 y$$

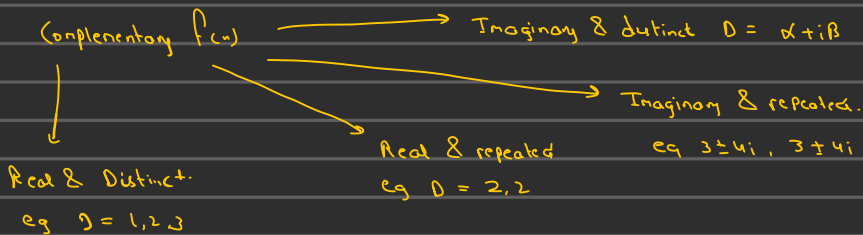
Auxiliary equation.

$$\frac{d^3 y}{dx^3} - 6 \frac{d^2 y}{dx^2} + 11 \frac{d y}{dx} - 6 y = 0$$

$$D^3 y - 6D^2 y + 11D y - 6y = 0$$

$$y(D^3 - 6D^2 + 11D - 6) = 0$$

$$D^3 - 6D^2 + 11D - 6 = 0$$



Type ① Real & different

$$y_1 = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + c_4 e^{m_4 x}$$

eg $D = 1, 2, 3$

$$y = c_1 e^{1x} + c_2 e^{2x} + c_3 e^{3x}$$

Real & repeated

$D = 2, 2$ $y = (c_1 x + c_2) e^{2x}$

$D = 1, 2, 3, 3$ $y = c_1 e^{1x} + c_2 e^{2x} + (c_3 x + c_4) e^{3x}$

Imaginary & Different

$D = 2 \pm 5i$ $\alpha = 2, \beta = 5$

$$y = [c_1 \cos \beta x + c_2 \sin \beta x] e^{\alpha x}$$

$$y = e^{2x} [c_1 \cos 5x + c_2 \sin 5x]$$

Imaginary & repeated

$D = 5 \pm 6i, 5 \pm 6i$ $\alpha = 5, \beta = 6$

$$y = [(c_1 x + c_2) \cos \beta x + (c_3 x + c_4) \sin \beta x] e^{\alpha x}$$

$$\underline{\text{Solve}} \Rightarrow \frac{d^3 y}{dx^3} - 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0$$

$$D^3 y - 6D^2 y + 11Dy - 6y = 0$$

$$y(D^3 - 6D^2 + 11D - 6) = 0$$

$$D^3 - 6D^2 + 11D - 6 = 0$$

$$\boxed{D = 1, 3, 2}$$

$$\text{The solution is } y = C_1 e^{1x} + C_2 e^{3x} + C_3 e^{2x}.$$

$$\underline{\text{Solve}} \quad \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} + 8y = 0$$

$$D^2 y + 5Dy + 8y = 0$$

$$y(D^2 + 5D + 8) = 0$$

$$D^2 + 5D + 8 = 0$$

$$D = 1, 2 \text{ (2)} \rightarrow \text{repeat if isn't shown (i.e. there must be 3 values)}$$

$$y_1 = C_1 e^x + (C_2 x + C_3) e^{2x}.$$

$$\underline{\text{Solve}} \quad (D^2 + 2D + 5)y = 0$$

$$D^2 + 2D + 5 = 0$$

$$D = -1 \pm 2i \quad \alpha = -1 \quad \beta = 2;$$

$$y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x]$$

$$y = e^{-1x} [C_1 \cos 2x + C_2 \sin 2x]$$

Solve $\frac{d^3 y}{dx^3} + y = 0$

$$D^3 y + y = 0$$

$$y(D^3 + 1) = 0$$

$$D^3 + 1 = 0$$

$$D = -1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \alpha = \frac{1}{2} \quad \beta = \frac{\sqrt{3}}{2}$$

$$y = c_1 e^{-x} + e^{\frac{1}{2}x} \left[c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right]$$

(L2) Particular Integral

Type ① when $\boxed{x = e^{ax}}$

$$\text{Solve } \frac{d^3 y}{dx^3} + 3 \frac{d^2 y}{dx^2} + 4y = e^{3x}$$

1stn CF
CF + PI

always put $D=a$

$$D^3 y + 3D^2 y + 4y = e^{3x}$$

$$y(D^3 + 3D^2 + 4) = e^{3x}$$

Auxiliary eqn. is $D^3 - 3D^2 + 4 = 0$

$$D = -1, 2, 2$$

$$\boxed{C.F = C_1 e^{-x} + (C_2 x + C_3) e^{2x}}$$

$$P.I = \frac{1}{f(D)} \times e^{ax}$$

$$= \frac{1}{D^3 - 3D^2 + 4} \times e^{3x} \quad a=3$$

$$= \frac{1}{3^3 + 3(3)^2 + 4} \times e^{3x}$$

$$PI = \frac{1}{4} \times e^{3x} = \frac{e^{3x}}{4}$$

Solution $y = C_1 e^{-x} + (C_2 x + C_3) e^{2x} + \frac{e^{3x}}{4}$

Solve $(D^3 - 2D^2 - 5D + 6)y = e^{3x} + 8$

$$D^3 - 2D^2 - 5D + 6 = 0$$

$$D = -2, 3, 1$$

$$\boxed{CF = C_1 e^{-2x} + C_2 e^{3x} + C_3 e^x}$$

$$PI = \frac{1}{D^3 - 2D^2 - 5D + 6} \times (e^{3x} + 8)$$

$$= \frac{1}{D^3 - 2D^2 - 5D + 6} e^{3x} + \frac{1}{D^3 - 2D^2 - 5D + 6} \times 8 \times \underbrace{(e^{0x})}_{\Rightarrow 1} =$$

Take deriv. = $\frac{x}{3D^2 - 4D - 5} \times e^{3x} + \frac{1}{6} \times 8(1)$

(uz the

dr=0

$$= \frac{x}{10} \times e^{3x} + \frac{1}{6} \times 8$$

$$= \frac{x}{10} e^{3x} + \frac{4}{3}$$

$$y = CF + PI.$$

$$PI = \frac{x e^{3x}}{10} + \frac{4}{3}$$

Solve $\frac{d^3 y}{dx^3} - 4 \frac{dy}{dx} = 2 \cosh 2x$

$$D^3 y - 4Dy = 2 \left(\frac{e^{2x} + e^{-2x}}{2} \right)^2$$

$$\cosh 2x = \frac{e^{2x} + e^{-2x}}{2}$$

$$(D^3 - 4D)y = 2 \left(\frac{e^{2x} + e^{-2x}}{4} \right)^2$$

$$(D^3 - 4D)y = \int_2^x ((e^{2u})^2 + 2(e^{2u})(e^{-2u}) + (e^{-2u})^2)$$

$$(D^3 - 4D)y = \int_2^x [e^{4u} + 2 + e^{-4u}]$$

$$[D^3 - 4D]y = \frac{e^{4x}}{2} + 1 + \frac{e^{-4x}}{2}$$

$$CF = ?$$

$$D^3 - 4D = 0$$

$$P.I = \frac{1}{D^3 - 4D} \times \left(\frac{e^{4x}}{2} + 1 + \frac{e^{-4x}}{2} \right)$$

$$\boxed{D =}$$

$$= \frac{1}{D^3 - 4D} \times \frac{e^{4x}}{2} + \frac{1}{D^3 - 4D} \times 1 + \frac{1}{D^3 - 4D} \times \frac{e^{-4x}}{2}$$

$$CF =$$

$$= \frac{1}{48} \frac{e^{4x}}{2} + \frac{x}{20^2 - 4} e^{0x} + \frac{1}{-48} \frac{e^{-4x}}{2}$$

$$= \frac{1}{48} \frac{e^{4x}}{2} + \frac{x}{-4} - \frac{1}{4} \frac{e^{-4x}}{2}$$

$$= -\frac{x}{4} + \frac{1}{48} \left[\frac{e^{4x}}{2} - \frac{e^{-4x}}{2} \right]$$

$$= -\frac{x}{4} + \frac{1}{48} \left[\frac{e^{4x} - e^{-4x}}{2} \right]$$

$$= -\frac{x}{4} + \frac{1}{48}$$

$$= -\frac{x}{4} + \frac{1}{48} \sinh 4x$$

$$y = CF + PI$$

(13) Particular Integral

Type - 2 (Tri) $D^2 = -a^2$

Q) Solve find P.I. of $(D^2 - 4D + 4)y = e^x + \cos 2x$

$$P.I. = \frac{1}{D^2 - 4D + 4} \times (e^x + \cos 2x)$$

$$= \frac{1}{D^2 - 4D + 4} \times e^x + \frac{1}{D^2 - 4D + 4} \times \cos 2x$$

$$= \frac{1}{1^2 - 4(1) + 4} \times e^x + \frac{1}{-4 - 4D + 4} \cos 2x$$

$$= \frac{1}{1} e^x + \frac{1}{-4D} \cos 2x$$

$$= e^x - \frac{1}{4D} \cos 2x$$

$$= e^x - \frac{1}{4} \frac{1}{D} \cos 2x$$

$$= e^x - \frac{1}{4} \int \cos 2x$$

$$= e^x - \frac{1}{4} \frac{(\sin 2x)}{2}$$

$$\boxed{P.I. = e^x - \frac{\sin 2x}{8}}$$

Q) Solve $(D^2 + 4)y = \cos 2x$

$y = CF + P.I.$

$$D^2 + 4 = 0 \quad D = \sqrt{-4}$$

$$D^2 = -4 \quad D = \pm 2i$$

$$CF = e^{dx} [C_1 \cos px + C_2 \sin px]$$

$$CF = e^{0n} [C_1 \cos 2n + C_2 \sin 2n]$$

$$CF = 1 [C_1 \cos 2n + C_2 \sin 2n]$$

$$P.I = \frac{1}{D^2 + 4} \times \cos 2n$$

$$= \frac{1}{-4+4} \rightarrow 0 \text{ Apply Derivative}$$

$$= \frac{x}{2D+0} \cos 2n$$

$$= \frac{x}{2} \int \cos 2n$$

$$= \frac{x}{2} \frac{\sin 2n}{2}$$

$$= \frac{x \sin 2n}{4}$$

$$y = CF + PI$$

$$= C_1 \cos 2n + C_2 \sin 2n + \frac{x \sin 2n}{2}$$

$$Q1) \text{ Solve } (D^2 - 5D + 6)y = \sin 3n$$

$$\underline{CF} \quad D^2 - 5D + 6 = 0$$

$$D = 3, 2$$

$$CF = C_1 e^{3n} + C_2 e^{2n}$$

$$P.I = \frac{1}{D^2 - 5D + 6} \cdot \sin 3n$$

$$P \cdot I = \frac{1}{-9 - 5D + 6}$$

$$P \cdot I = \frac{1}{-3 - 5D} \sin 3x$$

$$= \frac{1}{-3 - 5D} \frac{-3 + 5D}{-3 + 5D} \times \sin 3x$$

$$= \frac{-3 + 5D}{(-3)^2 - (5D)^2} \times \sin 3x$$

$$= \frac{-3 + 5D}{9 - 25D^2} \times \sin 3x$$

$$= \frac{-3 + 5D}{9 - 25(-9)} \times \sin 3x$$

$$= \frac{-3 + 5D}{234} \times \sin 3x$$

$$= \frac{1}{234} \int [-3 \sin 3x + 5D \sin 3x]$$

$$= \frac{1}{234} \left[-3 \sin 3x + 5 \frac{d}{dx} \sin 3x \right]$$

$$= \frac{1}{234} \left[-3 \sin 3x + 5 \times \cos 3x \times 3 \right]$$

$$= \frac{1 \times 3}{234} \left[-\sin 3x + 5 \cos 3x \right]$$

$$= \frac{1}{78} \left[5 \cos 3x - \sin 3x \right]$$

Type - ③ x^n

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

Q) Solve $(D^3 - 3D + 2)y = 1$

$$D^3 - 3D + 2 = 0$$

$$D = -2, 1, 1$$

$$CF = c_1 e^{-2x} + (c_2 + c_3 x) e^x$$

$$PI = \frac{1}{D^3 - 3D + 2} x^n$$

$$= \frac{1}{2 \left(\frac{D^2}{2} - \frac{3D}{2} + 1 \right)} x^n$$

$$PI = \frac{1}{2 \left(1 + \frac{D^2}{2} - \frac{3D}{2} \right)} x x$$

$$= \frac{1}{2 \left(1 + \frac{D^2 - 3D}{2} \right)} x x$$

$$= \frac{1}{2} \left[1 + \left(\frac{D^2 - 3D}{2} \right) \right]^{-1} x x$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$= \frac{1}{2} \left[1 - \frac{D^2 - 3D}{2} + \left(\frac{D^2 - 3D}{2} \right)^2 - \dots \right] x x$$

$$= \frac{1}{2} \left[x - \frac{(0-3)}{2} \right]$$

$$= \frac{1}{2} \left[x - \frac{-3}{2} \right]$$

$$= \frac{1}{2} \left[x + \frac{3}{2} \right]$$

$$\boxed{y = CF + PI}$$

Q) Solve $\frac{d^3 y}{dn^3} - \frac{2 dy}{dn} + 4y = 3n^2 - 5n + 2$

$$D^3 y - 2Dy + 4y = 3n^2 - 5n + 2$$

$$(D^3 - 2D + 4)y = 3n^2 - 5n + 2$$

$$D^3 - 2D + 4 = 0$$

$$D = -2, \pm i$$

$$C.F = e^{-2n} + e^{in} [C_1 \cos n + C_2 \sin n]$$

$$P.I = \frac{1}{D^3 - 2D + 4} [3n^2 - 5n + 2]$$

$$= \frac{1}{D^3 - 2D + 4} [3n^2 - 5n + 2]$$

$$4 \left[\frac{D^3}{4} - \frac{2D}{4} + 1 \right]$$

$$= \frac{1}{4 \left[1 + \frac{D^3}{4} - \frac{2D}{4} \right]} \times [3n^2 - 5n + 2]$$

$$= \frac{1}{4 \left(1 + \left(\frac{0^3 - 2 \cdot 0}{4} \right) \right)} \times [3n^2 - 5n + 2]$$

$$= \frac{1}{4} \left[1 + \left(\frac{0^3 - 2 \cdot 0}{4} \right) \right]^{-1} \times [3n^2 - 5n + 2]$$

$$(1+x)^{-1} = 1 - x + x^2 - \dots$$

$$= \left[1 - \left(\frac{0^3 - 2 \cdot 0}{4} \right) + \left(\frac{0^3 - 2 \cdot 0}{4} \right)^2 \right] \times (3n^2 - 5n + 2)$$

$$= \frac{1}{4} \left[3n^2 - 5n + 2 - \left(\frac{-2(0n - 5)}{4} \right) + \frac{4 \cdot 6}{16} \right]$$

$$= \frac{1}{4} \left[3n^2 - 5n + 2 - \left(\frac{-12n + 10}{4} \right) + \frac{3}{2} \right]$$

$$= \frac{1}{4} \left[3n^2 - 5n + 2 + \frac{12n - 10}{4} + \frac{3}{2} \right]$$

$$= \frac{1}{4} \left[3n^2 - 5n + 2 + \frac{12n}{4} - \frac{10}{4} + \frac{3}{2} \right]$$

$$= \frac{1}{4} [3n^2 - 2n + 1]$$

$$= \frac{3n^2}{4} - \frac{2n}{4} + \frac{1}{4}$$

L4 Particular Integral

Solve $(D^2 - 3D + 2)y = x^2 \cdot e^{2x}$

CF $D^2 - 3D + 2 = 0$

$D = 2, 1$

CF = $C_1 e^{2x} + C_2 e^x$

PI = $\frac{1}{D^2 - 3D + 2} x^2 \cdot e^{2x}$

that's right
there comes from e^{2x} if $\cdot 2x$

$= e^{2x} \frac{1}{(D+2)^2 - 3(D+2) + 2} x^2$

then $(D-2)^2 = e^{2x} \frac{1}{D^2 - 2(D)(2) + 2^2 - 3D - 6 + 2} x^2$

$= e^{2x} \frac{1}{D^2 + 4D + 4 - 3D - 6 + 2} x^2$

$= e^{2x} \frac{1}{D^2 + D} x^2$

$= e^{2x} \frac{1}{D(D+1)} x^2$

$= e^{2x} \frac{1}{D(1+D)} x^2$

$= e^{2x} \frac{1}{D} (1+D)^{-1} x^2$

$= e^{2x} \frac{1}{D} [1 - D + D^2] x^2$

$$D(x^2) = 2x$$

$$D^2(x^2) = 2$$

$$D^3(x^2) = 0$$

$$= e^{2x} \int \left[x^2 - 2x + 2 \right]$$

$$= e^{2x} \int (x^2 - 2x + 2) \cdot dx$$

$$= e^{2x} \left[\frac{x^3}{3} - \frac{2x^2}{2} + 2x \right]$$

$$= e^{2x} \left[\frac{x^3}{3} - x^2 + 2x \right]$$

$$y = CF + PI$$

$$Q) \text{ Find PI of } (D^2 + 2)y = x^2 e^{ix} + e^x - \cos 2x$$

$$(PI)_1 = \frac{1}{D^2 + 2} x^2 \cdot e^{2x}$$

$$= e^{2x} \frac{1}{(D+3)^2 + 2} x^2$$

$$= e^{2x} \frac{1}{D^2 + 6D + 9 + 2} x^2$$

$$= e^{2x} \frac{1}{D^2 + 6D + 11} x^2$$

$$(PI)_1 = e^{2x} \frac{1}{11 \left(\frac{D^2}{11} + \frac{6D}{11} + 1 \right)} x^2$$

$$= e^{2x} \frac{1}{11 \left(1 + \frac{D^2}{11} + \frac{6D}{11} \right)} x^2$$

$$= e^{2x} \frac{1}{11 \left(1 + \frac{D^2 + 6D}{11} \right)} x^2$$

$$= e^{2n} \frac{1}{11 \left[1 + \left(\frac{D^2 + 60}{11} \right) \right]^4} x^2$$

$$= \frac{e^{3n}}{11} \left[1 + \left(\frac{D^2 + 60}{11} \right) \right]^{-4} x^2$$

$$= \frac{e^{3n}}{11} \left[1 - \left(\frac{D^2 + 60}{11} \right) + \left(\frac{D^2 + 60}{11} \right)^2 \right] x^2$$

$$= \frac{e^{3n}}{11} \left[x^2 - \left(\frac{2 + 6(2n)}{11} \right) + \left(\frac{36(2)}{121} \right) \right]$$

$$= \frac{e^{3n}}{11} \left[x^2 - \left(\frac{2 + (2n)}{11} \right) + \frac{72}{121} \right]$$

$$= \frac{e^{3n}}{11} \left[x^2 - \frac{2}{11} - \frac{12n}{11} + \frac{72}{121} \right]$$

$$= \frac{e^{3n}}{11} \left[x^2 - \frac{12n}{11} + \frac{50}{121} \right]$$

$$(PJ)_x = \frac{1}{D^2 + 2} e^x$$

$$= \frac{1}{1^2 + 2} e^x = \frac{e^x}{3}$$

$$(PI)_3 = \frac{1}{D^2 + 2} (-\cos 2n)$$

$$D^2 = -0^2$$

$$= \frac{-1}{-2} \cos 2n$$

$$= \frac{\cos 2n}{2}$$

(15) Particular Integral

Type V

$$X = x^a \rightarrow \text{Tril Exp.}$$

$$\frac{1}{f(D)} x^a = \left\{ x - \frac{1}{f(D)} \times f'(D) \right\} \times \frac{1}{f(D)} x^a$$

Q) Solve $(D^2 + 4) = x \sin x$

$$CF = D^2 + 4 = 0$$

$$D = 2i, -2i$$

$$D = \alpha + \beta i$$

$$C.F = e^{2ix} [C_1 \cos 2x + C_2 \sin 2x]$$

$$CF = C_1 \cos 2x + C_2 \sin 2x$$

$$PI = \frac{1}{D^2 + 4} x \sin x$$

$$= \left\{ x - \frac{1}{D^2 + 4} \cdot 2D \right\} \frac{1}{D^2 + 4} x \sin x$$

$$= \left\{ x - \frac{1}{D^2 + 4} \cdot 2D \right\} \frac{1}{-1 + 4} \sin x$$

$$= \left\{ x - \frac{1}{D^2 + 4} \cdot 2 \cdot D \right\} \frac{\sin x}{3}$$

$$= \frac{x \sin x}{3} - \frac{1}{D^2 + 4} \cdot \frac{2 \cos x}{3}$$

$$= \frac{x \sin x}{3} - \frac{1}{3} \frac{1}{D^2+4} \cos x$$

$$= \frac{x \sin x}{3} - \frac{2}{3} \frac{1}{-1+4} \cos x$$

$$= \frac{x \sin x}{3} - \frac{2}{3} \times \frac{1}{3} \cos x$$

$$PI = \frac{x \sin x}{3} - \frac{2 \cos x}{9}$$

$$y = CF + PI.$$

Q) Solve $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = x \cdot e^x \sin x$

$$D^2 y - 2Dy + y = x \cdot e^x \sin x$$

$$(D^2 - 2D + 1)y = e^x \cdot x \sin x.$$

$$CF \quad D^2 - 2D + 1 = 0$$

$$CF =$$

$$PI = \frac{1}{D^2 - 2D + 1} e^x \cdot x \sin x$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 1} x \sin x$$

$$= e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 + 1}$$

$$= e^x \frac{1}{D^2} x \sin x \longrightarrow x \cdot v$$

$$= e^x \left[x - \frac{1}{D^2} \times 2D \right] \frac{1}{D^2} \times \sin x.$$

$$= e^x \left[x - \frac{1}{D^2} \times 2D \right] \times \frac{1}{-1} \sin x$$

$$= -e^x \left[x - \frac{1}{D^2} \times 2D \right] \times \sin x.$$

$$= e^x \left[x \sin x - \frac{1}{D^2} \times 2 \cos x \right]$$

$$= -e^x \left[x \sin x - \frac{2}{-1} \times \cos x \right]$$

$$\underline{\underline{PI}} = -e^x [x \sin x + 2 \cos x] \quad Y = CF + PI$$

when X not belongs to any type.

$$\frac{1}{D-a} x = e^{ax} \int e^{-ax} \cdot x \cdot dx$$

Q) Solve $(D^2 + 3D + 2)y = \sin e^x$

$$D^2 + 3D + 2 = 0$$

$$D = -1, -2$$

$$CF = C_1 e^{-x} + C_2 e^{-2x}$$

$$PI = \frac{1}{D^2 + 3D + 2} \sin e^x$$

$$= \frac{1}{(D+2)(D+1)} \sin e^x$$

$$= \frac{1}{(D+2)(D+1)} = \frac{A}{D+2} + \frac{B}{D+1}$$

$$\frac{1}{\cancel{(D+2)}(\cancel{D+1})} = \frac{A(\cancel{D+2}) + B(\cancel{D+1})}{\cancel{(D+2)}(\cancel{D+1})}$$

$$1 = A(D+1) + B(D+2)$$

$$\text{Put } D = -1$$

$$1 = B(-1+2)$$

$$1 = B(1)$$

$$\boxed{1 = B}$$

$$\text{Put } D = -2$$

$$1 = A(-2+1)$$

$$\boxed{-1 = A}$$

$$= \left[\frac{-1}{D+2} + \frac{1}{D+1} \right] \text{line}^x$$

$$= \frac{-1}{D+2} \text{line}^x + \frac{1}{D+1} \text{line}^x$$

$$= -1 \left[e^{-2x} \int e^{2x} \cdot \text{line}^x \cdot dx \right] + \left[e^{-x} \int e^x \cdot \text{line}^x \cdot dx \right]$$

$$= - \left[\frac{1}{e^{2x}} \int e^x \cdot e^x \cdot \text{line}^x \cdot dx \right] + \left[\frac{1}{e^x} \int e^x \cdot \text{line}^x \cdot dx \right]$$

$$\text{Put } e^x = t$$

$$e^x = \frac{dt}{dx}$$

$$\boxed{e^x dx = dt}$$

$$= - \left[\frac{1}{(t)^2} \int t \cdot \text{line}^t \cdot dt \right] + \left[\frac{1}{t} \int \text{line}^t \cdot dt \right]$$

$$= - \left[\frac{1}{t^2} \left(t \cdot (-\cos t) - 1(-\sin t) \right) + \left[\frac{1}{t} (-\cos t) \right] \right]$$

$$= - \left[\frac{1}{t^2} (-t \cos t + \sin t) \right] - \frac{\cos t}{t}$$

$$= - \left[- \frac{\cos t}{t} + \frac{\sin t}{t^2} \right] - \frac{\cos t}{t}$$

$$= \cancel{\frac{\cos t}{t}} - \frac{\sin t}{t^2} - \cancel{\frac{\cos t}{t}}$$

$$= - \frac{\sin t}{t^2} = - \frac{\sin e^x}{(e^x)^2} = - \frac{\sin e^x}{e^{2x}} = - e^{-2x} \sin e^x = \text{PI}$$

Q) Solve $D^2 y + 3Dy + 2y = e^{e^x}$

$$(D^2 + 3D + 2)y = e^{e^x}$$

$$D^2 + 3D + 2 = 0$$

$$D = -1, -2$$

$$\boxed{\text{CF} = C_1 e^{-x} + C_2 e^{-2x}}$$

$$\text{PI} = \frac{1}{D^2 + 3D + 2} e^{e^x}$$

$$= \frac{1}{(D+2)(D+1)} e^{e^x}$$

$$= \frac{1}{D+2} \left[e^{-x} \int e^x e^{e^x} \cdot dx \right] - \textcircled{I}$$

Find $\int e^x e^{e^x} \cdot dx$ Put $e^x = t$ $e^x = \frac{dt}{dx}$ $e^x dx = dt$

$$\int e^t \cdot dt$$

$$I = e^t = e^{e^x} - (2)$$

$$\text{from (1)} \quad PI = \frac{1}{(0+2)} \left[e^{-x} \cdot e^{e^x} \right]$$

$$= \left[e^{-2x} \int e^{2x} \cdot e^{-x} \cdot e^{e^x} \cdot dx \right]$$

$$= e^{-2x} \int e^x e^{e^x} \cdot dx$$

$$\boxed{PI = e^{-2x} e^{e^x}}$$

① Method of variation of Parameter

Step ① $CF = C_1 y_1 + C_2 y_2$

$$y_1 =$$

$$y_2 =$$

Step ② $\omega = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

Step ③ $u = - \int \frac{u_2 x}{\omega} dx \quad v = \int \frac{y_1 x}{\omega} dx$

Step ④ $PI = uy_1 + vy_2$

Q.) Solve by the method of variation of parameter

$$\frac{d^2 y}{dx^2} + y = \sec x \cdot \tan x$$

$$D^2 y + y = \sec x \cdot \tan x$$

$$(D^2 + 1)y = \sec x \cdot \tan x$$

$$CF = D^2 + 1 = 0$$

$$D = i, -i \quad D = 0 \pm i$$

$$CF = C_1 \cos x + C_2 \sin x$$

$$y_1 = \cos x \quad y_2 = \sin x \quad x = \sec x \cdot \tan x$$

$$w = \begin{vmatrix} \cos u & \sin u \\ -\sin u & \cos u \end{vmatrix} = \cos^2 u - (-\sin^2 u) \\ = 1 = w$$

$$u = - \int \frac{y_2 \cdot x}{w} \cdot du$$

$$u = - \int \frac{\sin u (\sec u \cdot \tan u)}{1} \cdot du$$

$$u = - \int \sin u \cdot \frac{1}{\cos u} \cdot \frac{\sin u}{\cos u} \cdot du$$

$$u = - \int \frac{\sin^2 u}{\cos^2 u} \cdot du$$

$$u = \int \tan^2 u \cdot du$$

$$u = - \int (\sec^2 u - 1) \cdot du$$

$$u = -\tan u - u$$

$$u = -\tan u + u$$

$$v = \int \frac{y_1 \cdot x}{w} \cdot du$$

$$= \int \frac{\cos u (\sec u \cdot \tan u)}{1} \cdot du$$

$$= \int \frac{\cancel{\cos u} \cdot 1}{\cancel{\cos u}} \cdot \tan u \cdot du$$

$$= \int \tan u \cdot du$$

$$\boxed{v = \log |\sec u|}$$

$$PI = uy_1 + vy_2$$

$$\boxed{PI = (-\tan u + u) \cos u + \log |\sec u| \sin u}$$

$$y = CF + PI$$

$$y = \bigcirc + \bigcirc$$

Q.2) Solve $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x}$

$$D^2y + 3Dy + 2y = e^{e^x}$$

$$(D^2 + 3D + 2)y = e^{e^x}$$

$$D^2 + 3D + 2 = 0$$

$$D = -1, -2$$

$$CF = C_1 e^{-x} + C_2 e^{-2x}$$

$$y_1 = e^{-x} \quad y_2 = e^{-2x} \quad x = e^{e^x}$$

$$\begin{aligned} w &= \begin{vmatrix} e^{-x} & e^{-2x} \\ -e^{-x} & -2e^{-2x} \end{vmatrix} = -2e^{-x}e^{-2x} + e^{-x}e^{-3x} \\ &= -2e^{-3x} + 1e^{-3x} \\ &= -1e^{-3x} \\ &= -e^{-3x} \end{aligned}$$

$$u = - \int \frac{y_2 x}{w} \cdot dx = - \int \frac{e^{-2x} e^{e^x}}{-e^{-3x}} \cdot dx$$

$$u = \int e^x \cdot e^{e^x} \cdot dx$$

$$\text{Put } e^x = t$$

$$e^x = \frac{dt}{dx} \quad e^x \cdot dx = dt$$

$$u = \int e^t \cdot dt = e^t$$

$$u = e^{e^x}$$

$$u = \int \frac{y_1 x}{w} \cdot du$$

$$= \int \frac{e^{-x} e^{e^x}}{-e^{-2x}} \cdot du$$

$$= - \int e^{2x} e^{e^x} \cdot du$$

$$= - \int e^x \cdot e^x \cdot e^{e^x} \cdot du$$

put $e^x = t$ $e^x = \frac{du}{dt}$ $e^x \cdot du = dt$.

$$P \cdot J = u y_1 + v y_2$$

$$= e^{e^x} e^{-x} + (e^x e^{e^x} - e^{e^x}) \cdot e^{-2x}$$

$$= e^{e^x} e^{-x} - e^x e^{e^x} e^{-2x} + e^{e^x} e^{-2x}$$

$$= e^{e^x} e^{-x} - e^{-x} e^{e^x} + e^{-2x} \cdot e^{e^x}$$

$$P1 = e^{-2x} \cdot e^{e^x}$$

$$= \int t \cdot e^t \cdot dt$$

$$= \{ t \cdot e^t - e^t \}$$

$$= e^x \cdot e^{e^x} - e^{e^x}$$

$$= e^{e^x} (e^x - 1)$$

$$v = e^{e^x} (e^x - 1)$$

Mod-③ Beta & gamma f(n)

① Property of gamma f(n)

$$\int_0^{\infty} e^{-x} x^{n-1} \cdot dx = \Gamma n$$

$$\int_0^{\infty} e^{-x} x^n \cdot dx = \Gamma_{n+1}$$

$$\begin{aligned}\Gamma_0 &= \infty \\ \Gamma_1 &= 1 \\ \Gamma_{\frac{1}{2}} &= \sqrt{\pi}\end{aligned}$$

$$\begin{aligned}\Gamma n &= (n-1)! & \text{eg:- } \Gamma_5 &= 4! \\ \Gamma n &= \frac{\Gamma_{n+1}}{n} \\ \Gamma n &= (n-1) \Gamma_{(n-1)}\end{aligned}$$

find Γn

i) if n is a true integer

$$\text{eg } \Gamma_5 = 4! = 4 \times 3 \times 2 \times 1 = 24$$

$$\Gamma_6 = 5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$\Gamma_7 = 6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

ii) if n is a negative integer (always answer $= \infty$)

$$\text{eg } \Gamma_{-4} = \text{by using } \frac{\Gamma_{n+1}}{n} = \frac{\Gamma_{-4+1}}{-4} = \frac{\Gamma_{-3}}{-4} = \frac{1}{-4} \frac{\Gamma_{-3+1}}{(-3)}$$

$$= \frac{\Gamma_{-2}}{12} = \frac{\Gamma_{-2+1}}{12 \times (-2)} = \frac{\Gamma_{-1}}{-24} = \frac{\Gamma_{-1+1}}{-24 \times (-1)} = \infty$$

iii) If n is positive fraction :-

$$\begin{aligned}
 \text{eg } \sqrt{\frac{5}{2}} &= \left(\frac{5}{2} - 1\right) \sqrt{\frac{5}{2} - 1} \\
 &= \frac{3}{2} \sqrt{\frac{3}{2}} = \frac{3}{2} \left(\frac{3}{2} - 1\right) \sqrt{\frac{3}{2} - 1} \\
 &= \frac{3}{2} \times \frac{1}{2} \sqrt{\frac{1}{2}} \\
 &= \frac{3}{4} \sqrt{\frac{1}{2}}
 \end{aligned}$$

iv) If n is negative fraction

$$\begin{aligned}
 \text{eg } \sqrt{\frac{-5}{3}} &= \frac{\sqrt{\frac{-5}{3} + 1}}{\frac{-5}{3}} = \frac{\sqrt{\frac{-2}{3}}}{\frac{-5}{3}} = \frac{\sqrt{\frac{-2}{3} + 1}}{\left(\frac{-5}{3}\right)\left(\frac{-2}{3}\right)} \\
 &= \frac{\sqrt{\frac{1}{3}}}{\frac{10}{9}}
 \end{aligned}$$

② Numericals on types of Gamma function

Type - I

$$\int_0^{\infty} e^{-ax^n} dx = \frac{1}{n \cdot n! a} \sqrt[n]{n}$$

eg $\int_0^{\infty} e^{-x^4} \cdot dx =$ Put $x^4 = t$
 $x = t^{1/4}$

$$\frac{dx}{dt} = \frac{1}{4} t^{-3/4}$$

$$dx = \frac{1}{4} t^{-3/4} \cdot dt$$

$$x = 0 \quad x = \infty$$

$$t = 0 \quad t = \infty$$

$$\int_0^{\infty} e^{-t} \frac{1}{4} t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \int_0^{\infty} e^{-t} t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \sqrt[n]{\frac{-3}{4} + 1} = \frac{1}{4} \sqrt[n]{\frac{1}{4}}$$

Type - II

$$\int_0^{\infty} x^n e^{-ax^n} \cdot dx = \frac{1}{n} \frac{1}{a^{n+1/n}} \sqrt[n]{n+1}$$

eg $\int_0^{\infty} x \cdot e^{-x^4} \cdot dx$

Let $x^4 = t$

$x = t^{1/4}$

$\frac{dx}{dt} = \frac{1}{4} t^{-3/4}$

$dx = \frac{1}{4} t^{-3/4} \cdot dt$

$x = 0$

$t = 0$

$x = \infty$

$t = \infty$

$= \int_0^{\infty} t^{1/4} e^{-t} \frac{1}{4} t^{-3/4} \cdot dt$

$= \frac{1}{4} \int_0^{\infty} e^{-t} t^{\frac{1-3}{4}} \cdot dt$

$= \frac{1}{4} \int_0^{\infty} e^{-t} t^{-2/4} \cdot dt = \frac{1}{4} \int_0^{\infty} e^{-t} t^{-1/2} \cdot dt$

$= \frac{1}{4} \left| \frac{-1}{\frac{1}{2}} + 1 \right| = \frac{1}{4} \left| \frac{1}{2} \right| = \frac{1}{4} \sqrt{\pi}$

Q) ST $\underbrace{\int_0^{\infty} x e^{-x^8} \cdot dx}_{I_1} \cdot \underbrace{\int_0^{\infty} x^2 \cdot e^{-x^4} \cdot dx}_{I_2} = \frac{\pi}{16\sqrt{2}}$

$I_1 = \int_0^{\infty} x e^{-x^8} \cdot dx$

$I_2 = \int_0^{\infty} x^2 e^{-x^4} \cdot dx$

put $x^8 = t$

$x = t^{1/8}$

put $x^4 = t$ $x = t^{1/4}$

$$\frac{du}{dt} = \frac{1}{8} t^{-2/8}$$

$$du = \frac{1}{8} t^{-2/8} \cdot dt$$

$$I_1 = \int_0^{\infty} t^{1/8} e^{-t} \frac{1}{8} t^{-2/8} dt$$

$$I_1 = \frac{1}{8} \int_0^{\infty} e^{-t} t^{-1/4} \cdot dt$$

$$I_1 = \frac{1}{8} \left| \frac{-3/4 + 1}{-3/4} \right| = \frac{1}{8} \left| \frac{1}{-3/4} \right|$$

$$\frac{dv}{dt} = \frac{1}{4} t^{-3/4}$$

$$dv = \frac{1}{4} t^{-3/4} \cdot dt$$

$$I_2 = \int_0^{\infty} (t^{1/4})^2 e^{-t} \frac{1}{4} t^{-3/4} dt$$

$$I_2 = \frac{1}{4} \int_0^{\infty} t^{2/4} e^{-t} t^{-3/4} dt$$

$$= \frac{1}{4} \int_0^{\infty} e^{-t} t^{-1/4} \cdot dt$$

$$= \frac{1}{4} \left| \frac{-1/4 + 1}{-1/4} \right| = \frac{1}{4} \left| \frac{3}{-1/4} \right|$$

$$I_1 \cdot I_2 = \frac{1}{8} \left| \frac{1}{-3/4} \right| \times \frac{1}{4} \left| \frac{3}{-1/4} \right|$$

$$= \frac{1}{32} \left| \frac{1}{-3/4} \right| \cdot \left| \frac{3}{-1/4} \right|$$

$$= \frac{1}{32} (\sqrt{2\pi})$$

$$= \frac{\sqrt{2}}{32} \pi \times \frac{\sqrt{2}}{2} = \frac{2\pi}{32 \sqrt{2}} = \boxed{\frac{\pi}{16 \sqrt{2}}}$$

Type ③ $\int \log x = -1$

$$\int_0^1 x^n (\log x)^n \cdot du = \frac{(-1)^n}{(n+1)^{n+1}} \overline{1^{n+1}}$$

evaluate $= \int_0^1 (x \log x)^4 \cdot du$

$$= \int_0^1 x^u (\log x)^4 \cdot dx$$

$$\text{put } \log_t x = -t$$

$$x = e^{-t}$$

$$\frac{dx}{dt} = e^{-t} (-1)$$

$$\boxed{dx = -e^{-t} \cdot dt}$$

$$x=0 \quad t=\infty$$

$$x=1 \quad t=0$$

$$= \int_0^1 (e^{-t})^u (-t)^4 (-e^{-t} dt)$$

$$= - \int_0^1 e^{-4t} t^4 e^{-t} \cdot dt$$

$$= \int_0^{\infty} e^{-5t} \cdot t^4 \cdot dt$$

$$\text{put } 5t = u$$

$$5 \frac{dt}{du} = 1$$

$$5 dt = du$$

$$\boxed{dt = \frac{1}{5} \cdot du}$$

$$t=0 \quad u=0$$

$$t=\infty \quad u=\infty$$

$$= \int_0^{\infty} e^{-u} \left(\frac{u}{5}\right)^4 \cdot \frac{1}{5} \cdot du$$

$$= \frac{1}{5} \frac{1}{5^4} \int_0^{\infty} e^{-u} u^4 \cdot du$$

$$= \frac{1}{5^5} \sqrt{4+1}$$

$$= \frac{1}{5^5} \sqrt{5} = \frac{1}{5^5} \cdot 41 = \frac{24}{5^5}$$

Type IV

put $a^n = e^t$

$$\int_0^{\infty} \frac{x^a}{a^x} \cdot dx = \frac{1}{(\log a)^{a+1}} \sqrt{a+1}$$

evaluate $\int_0^{\infty} \frac{x^7}{2^x} \cdot dx$

$$x = \frac{t}{\log 2}$$

put $2^x = e^t$

$$\log_e 2^x = t$$

$$\frac{dx}{dt} = \frac{1}{\log 2} \times (1)$$

$$dx = \frac{1}{\log 2} \cdot dt$$

$$x \log 2 = t$$

$$x = 0$$

$$x = \infty$$

$$t = 0$$

$$t = \infty$$

$$\int_0^{\infty} \frac{\left(\frac{t}{\log 2}\right)^7}{e^t} \cdot \frac{1}{\log 2} \cdot dt$$

$$= \int_0^{\infty} e^{-t} \frac{t^7}{(\log 2)^7} \times \frac{1}{(\log 2)} \cdot dt$$

$$= \frac{1}{(\log 2)^8} \int_0^{\infty} e^{-t} \cdot t^7 \cdot dt$$

$$= \frac{1}{(\log 7)^8} \times \sqrt{7+1}$$

$$= \frac{1}{(\log 7)^8} \sqrt{8}$$

Solve $\int_0^{\infty} 4\sqrt{x} e^{-\sqrt{x}} \cdot dx$

$$= \int_0^{\infty} x^{1/4} e^{-\sqrt{x}} \cdot dt.$$

$$\sqrt{x} = t$$

$$x = t^2$$

$$\frac{dx}{dt} = 2t$$

$$\boxed{dx = 2t \cdot dt.}$$

$$= \int_0^{\infty} (t^2)^{1/4} e^{-t} 2t dt.$$

$$= 2 \int_0^{\infty} t^{1/2} e^{-t} t^1 dt.$$

$$= 2 \int_0^{\infty} e^{-t} t^{3/2} dt.$$

$$2 \sqrt{\frac{3}{2} + 1}$$

$$= 2 \sqrt{\frac{5}{2}}$$

Q.1) $\int_0^{\infty} \frac{dx}{3^{4x^2}}$

$$3^{4x^2} = e^t$$

$$\log_e 3^{4x^2} = 4$$

$$4x^2 \cdot \log 3 = 4$$

$$x^2 = \frac{1}{4 \log 3}$$

$$x = \frac{\sqrt{t}}{2\sqrt{\log 3}}$$

$$\frac{dx}{dt} = \frac{1}{2 \log 3} \cdot \frac{1}{2\sqrt{t}}$$

$$dx = \frac{1}{4\sqrt{\log 3}} \cdot \frac{dt}{\sqrt{t}}$$

$$= \int_0^{\infty} \frac{1}{e^t} \cdot \frac{1}{4\sqrt{\log 3}} \times \frac{1}{\sqrt{t}} \cdot dt$$

$$= \frac{1}{4\sqrt{\log 3}} \int_0^{\infty} e^{-t} \cdot t^{-1/2} \cdot dt$$

$$= \frac{1}{4\sqrt{\log 3}} \left[\frac{t^{-1/2+1}}{-1/2+1} \right]$$

$$= \frac{1}{4\sqrt{\log 3}} \times \left[\frac{1}{2} \right]$$

$$Q) \int_0^{\infty} x^a e^{-2x^2} \cdot dx$$

$$\text{put } 2x^2 = t$$

$$x^2 = \frac{t}{2}$$

$$x = \frac{\sqrt{t}}{\sqrt{2}}$$

$$\frac{du}{dt} = \frac{1}{\sqrt{t}} \times \frac{1}{2\sqrt{t}}.$$

$$du = \frac{1}{2\sqrt{t} \sqrt{t}} \cdot dt.$$

$$= \int_0^{\infty} \left(\frac{\sqrt{t}}{\sqrt{t}} \right)^9 e^{-t} \frac{1}{2\sqrt{t} \sqrt{t}} \cdot dt.$$

$$= \frac{1}{(\sqrt{t})^9 (\sqrt{t})} \int_0^{\infty} (\sqrt{t})^9 e^{-t} t^{-1/2} \cdot dt.$$

$$= \frac{1}{16 \sqrt{t} \times 2 \sqrt{t}} \times \int_0^{\infty} t^{9/2} e^{-t} t^{-1/2} \cdot dt.$$

$$= \frac{1}{64} \int_0^{\infty} e^{-t} t^4 \cdot dt.$$

$$= \frac{1}{64} \Gamma 5$$

③ Beta function & its types

$$\int_0^1 x^{n-1} (1-x)^{n-1} dx = \beta(n, n)$$

$$\int_0^1 x^m (1-x)^n dx = \beta(m+1, n+1) \quad \underline{\underline{\text{Imp}}}.$$

Properties of β fcn

$$1) \beta(m, n) = \beta(n, m)$$

$$2) \beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} = \frac{(m-1)! (n-1)!}{(m+n-1)!}$$

$$\beta(3, 2) = \frac{\Gamma 3 \Gamma 2}{\Gamma 5} = \frac{2! \times 1!}{4!} \Rightarrow$$

$$\frac{\Gamma p \Gamma(1-p)}{\sin p \pi}$$

Duplication formula

$$\Gamma n \Gamma n + \frac{1}{2} = \frac{\Gamma n \Gamma 2n}{2^{2n-1}}$$

$$\text{Find } \Gamma \frac{1}{2} \Gamma \frac{3}{4} = \Gamma \frac{1}{4} \Gamma \frac{1}{4} + \frac{1}{2}$$

$$\Gamma \frac{3}{4} \Gamma \frac{5}{4} = \frac{\pi}{2 \sqrt{2}}$$

$$\Gamma \frac{5}{4} \Gamma \frac{7}{4} = \frac{3 \sqrt{2} \pi}{16}$$

Type - I

$$\int_0^a x^m (a-x)^n \cdot dx = a^{m+n+1} \beta(m+1, n+1)$$

eg $\int_0^4 \sqrt{x} (4-x)^{3/2} \cdot dx.$

put $x = 4t$

$$\frac{dx}{dt} = 4$$

$$dx = 4dt.$$

$$x = 0 \quad t = 0$$

$$x = 4 \quad t = 1$$

$$\int_0^1 \sqrt{4t} (4-4t)^{3/2} \cdot 4 \cdot dt$$

$$= \int_0^1 2\sqrt{t} \cdot 4^{3/2} (1-t)^{3/2} \cdot 4 \cdot dt$$

$$= 2 \cdot 4^{3/2} \cdot 4 \int_0^1 t^{1/2} (1-t)^{3/2} \cdot dt.$$

$$= 2 \times 8 \times 4 \times \beta\left(\frac{1}{2}+1, \frac{3}{2}+1\right)$$

$$= 2 \times 8 \times 4 \times \beta\left(\frac{3}{2}, \frac{5}{2}\right)$$

$$= 64 \beta\left(\frac{3}{2}, \frac{5}{2}\right)$$

$$= 64 \frac{\sqrt{\frac{3}{2}} \sqrt{\frac{5}{2}}}{\sqrt{\frac{3}{2} + \frac{5}{2}}}$$

$$= 64 \frac{\frac{1}{2} \sqrt{\frac{1}{2}} \cdot \frac{3}{2} \sqrt{\frac{3}{2}}}{\sqrt{4}}$$

$$= 64 \times \frac{\frac{1}{2} \times \sqrt{2} \times \frac{3}{2} \cdot \frac{1}{2} \sqrt{2}}{\cancel{2 \times 2 \times 1}}$$

$$= \frac{64 \times 1 \times \sqrt{2} \times 1 \times \sqrt{2}}{\cancel{2 \times 2 \times 2 \times 2}} = 4\pi$$

Type - II

$$\int_0^1 x^n (1-x^n)^p \cdot dx = \frac{1}{n} B\left(\frac{n+1}{n}, p+1\right)$$

eg $\int_0^1 x^6 (1-x^2)^{1/2} \cdot dx$

put $x^2 = t$

$x = t^{1/2}$

$$\boxed{\frac{dx}{dt} = \frac{1}{2} t^{-1/2}}$$

$dx = \frac{1}{2} t^{-1/2} \cdot dt$

$x = 0 \quad t = 0$

$x = 1 \quad t = 1$

$$= \int_0^1 (t^{1/2})^6 (1-t)^{1/2} \cdot \frac{1}{2} t^{-1/2} \cdot dt$$

$$= \frac{1}{2} \int_0^1 t^{6/2} (1-t)^{1/2} t^{-1/2} \cdot dt$$

$$= \frac{1}{2} \int_0^1 t^{5/2} (1-t)^{1/2} \cdot dt.$$

$$= \frac{1}{2} P \left(\frac{5}{2} + 1, \frac{1}{2} + 1 \right)$$

$$= \frac{1}{2} P \left(\frac{7}{2}, \frac{3}{2} \right)$$

$$\sim \int_0^3 \frac{x^{3/2}}{\sqrt{3-x}} \cdot dx \quad \int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{432}{35} \pi$$

I_1

I_2

$$\text{Let } I_1 = \int_0^3 \frac{x^{3/2}}{\sqrt{3-x}} \cdot dx$$

$$I_2 = \int_0^1 \frac{du}{\sqrt{1-u^4}}$$

$$I_1 = \int_0^3 x^{3/2} (3-x)^{-1/2} \cdot dx.$$

$$I_2 = \int_0^1 (1-u^4)^{-1/2} \cdot du$$

$$\text{put } x=3t$$

$$\text{put } x^{1/4} = t.$$

$$u = t^4$$

$$\frac{dx}{dt} = 3$$

$$\frac{du}{dt} = 4t^3$$

$$\boxed{dx = 3dt}$$

$$\boxed{du = 4t^3}$$

$$x=0 \quad t=0$$

$$u=0 \quad t=0$$

$$x=3 \quad t=1$$

$$u=1 \quad t=1$$

$$I_1 = \int_0^1 (3t)^{3/2} (3-3t)^{-1/2} \cdot 3 \cdot dt.$$

$$= \int_0^1 (1-t)^{-1/2} 4 \cdot t^3 \cdot dt.$$

$$= 4 \int_0^1 t^3 (1-t)^{-1/2} \cdot dt$$

$$= 4 P \left(3+1, -\frac{1}{2}+1 \right)$$

$$I_1 = 3^{3/2} \cdot 3^{-1/2} \int_0^1 t^{3/2} (1-t)^{-1/2} \cdot dt.$$

$$I_1 = 3\sqrt{3} \times 3 \times \frac{1}{\sqrt{3}} \times \beta\left(\frac{3}{2}+1, -\frac{1}{2}+1\right) = 4\beta\left(4, \frac{1}{2}\right)$$

$$I_1 = 9 \times \beta\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$I_1 \times I_2 = 9 \times \frac{\sqrt{\frac{5}{2}} \sqrt{\frac{1}{2}}}{\sqrt{3}} \times 4 \times \frac{\sqrt{4} \times \sqrt{\frac{1}{2}}}{\sqrt{\frac{9}{2}}}$$

Type - III

$$\int_0^1 (1 - n\sqrt{n})^m dx = \frac{m!n!}{(m+n)!}$$

evaluate $\int_0^1 (1 - 3\sqrt{n})^{11/2} \cdot dn$

put $3\sqrt{n} = t$

$$n = t^3$$

$$\frac{dn}{dt} = 3t^2$$

$$dn = 3t^2 dt$$

$$\int_0^1 (1-t)^{11/2} 3t^2 \cdot dt = 3 \int_0^1 (1-t)^{11/2} t^2 dt$$

$$= 3 \int_0^1 t^2 (1-t)^{11/2} dt$$

$$= 3\beta\left(2+1, \frac{11}{2}+1\right)$$

$$= 3\beta\left(3, \frac{13}{2}\right)$$

④ Beta function & its types

- II

Type IV

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$\int_0^{\pi/2} \sin^4 x \cdot dx$$

$$\int_0^{\pi/2} \cos^4 x \cdot dx =$$

$$\int_0^{\pi/2} \sin^4 x \cdot dx = \frac{3}{2} \cdot \frac{1}{2} \times \frac{\pi}{2}$$

$$\int_0^{\pi/2} \cos^6 \theta \cdot d\theta = \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \times \frac{\pi}{2}$$

$$\int_0^{\pi/2} \cos^5 \theta \cdot dx = \frac{4}{5} \cdot \frac{2}{3}$$

$$\int_0^{\pi/2} \sin^m x \cdot \cos^n x \cdot dx = \frac{\overbrace{(n-1)(n-3)\dots}^{2 \text{ or } 1} \cdot \overbrace{(m-1)(m-3)\dots}^{2 \text{ or } 1}}{\overbrace{(n+1)(n+3)\dots}^{2 \text{ or } 1}} \cdot \frac{\pi}{2}$$

$$\int_0^{\pi/2} \sin^4 x \cdot \cos^5 x \cdot dx = \frac{3 \times 1 \times 4 \times 2}{4 \times 7 \times 5 \times 3 \times 1}$$

$$\int_0^a (a^2 - x^2)^{5/2} \cdot dx$$

Put $x = a \sin \theta$

$$\frac{dx}{d\theta} = a \cos \theta$$

$$\sin^{-1}\left(\frac{x}{a}\right) = \theta$$

$$dx = (a \cos \theta) \cdot d\theta$$

$$x = 0 \quad \theta = 0$$

$$x = a \quad \theta = \pi/2$$

$$\int_0^{\pi/2} (a^2 - a^2 \sin^2 \theta)^{5/2} \cdot a \cos \theta \cdot d\theta$$

$$= a \int_0^{\pi/2} (a^2)^{5/2} (1 - \sin^2 \theta)^{5/2} \cos \theta \cdot d\theta$$

$$= a \cdot a^5 \int_0^{\pi/2} (\cos^2 \theta)^{5/2} \cos \theta \cdot d\theta$$

$$= a^6 \int_0^{\pi/2} \cos^6 \theta \cdot d\theta$$

$$= a^6 \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \pi/2$$

evaluate $\int_0^1 x^5 \sin^{-1} x \cdot dx$

Type V

$$\int_0^{\infty} \frac{x^n}{(1+x)^m} dx = \frac{1}{2} B\left(\frac{n+1}{2}, \frac{2n-m-1}{2}\right)$$

$$\text{eg : } \int_0^{\infty} \frac{dx}{(1+x)^{7/2}}$$

$$\text{put } x = \tan \theta$$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$dx = \sec^2 \theta \cdot d\theta$$

$$x=0 \quad \theta=0$$

$$x=\infty \quad \theta = \pi/2$$

$$= \int_0^{\pi/2} \frac{1}{(1+\tan^2 \theta)^{7/2}} \sec^2 \theta \cdot d\theta$$

$$= \int_0^{\pi/2} \frac{1}{(\sec^2 \theta)^{7/2}} \sec^2 \theta \cdot d\theta$$

$$= \int_0^{\pi/2} \frac{1}{(\sec \theta)^7} \times \sec^2 \theta \cdot d\theta$$

$$= \int_0^{\pi/2} \frac{1}{\sec^5 \theta} \cdot d\theta$$

$$= \int_0^{\pi/2} \frac{1}{\sec^5 \theta} d\theta$$

$$= \int_0^{2\pi} \cos^2 \theta \cdot d\theta$$

$$= \frac{4}{3} \cdot \frac{2}{3}$$

Type VI

$$\int_0^{\infty} \frac{x^n}{(1+x)^n} = \beta(n+1, n-m-1)$$

$$\text{eg } \int_0^{\infty} \frac{x^5(1+x^4)}{(1+x)^6} \cdot dx = \int_0^{\infty} \frac{(x^5 + x^9 \cdot x^2)}{(1+x)^6} \cdot dx$$

$$= \int_0^{\infty} \frac{x^5 + x^9}{(1+x)^6} \cdot dx$$

$$= \int_0^{\infty} \frac{x^5}{(1+x)^6} \cdot dx + \int_0^{\infty} \frac{x^9}{(1+x)^6} \cdot dx$$

$$= \beta(6, 10) + \beta(10, 6)$$

$$= 2\beta(6, 10)$$

Type VII

$$\int_0^1 \frac{x^{n-1}(1-x)^{n-1}}{(a+x)^{m+n}} \cdot dx = \frac{\beta(n, n)}{(a+b)^m a^n}$$

Type VIII

$$\int_0^1 \frac{x^{n-1} x^{n-1}}{(1+x)^{m+n}} = \beta(n, n)$$

Type IX

$$\int_a^b (x-a)^n (b-x)^n \cdot dx = (b-a)^{2n+1} \beta(n+1, n+1)$$

eg $\int_7^{11} 4(x-7)(11-x) \cdot dx$

$$\int_7^{11} (x-7)^{1/4} (11-x)^{1/4} \cdot dx$$

$a=7, b=11, n=n=1/4$

$$= 4^{3/2} \beta\left(\frac{1}{4}+1, \frac{1}{4}+1\right)$$

$$= 2^{6 \times 3/2} \beta\left(\frac{5}{4}, \frac{5}{4}\right)$$

$$= 8 \beta\left(\frac{5}{4}, \frac{5}{4}\right)$$

Type X

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

Q) $\int_0^{\pi/2} \frac{dx}{\sqrt{1 - \frac{1}{2} \sin^2 \theta}}$

Put $\cos^2 \theta = t$
 $\cos^2 \theta = t^{1/2}$

$$\cos \theta = t^{1/4}$$

$$\theta = \cos^{-1} \left(t^{1/4} \right)$$

$$\cos^{-1} u = \frac{-1}{\sqrt{1-u^2}}$$

$$\frac{d\theta}{dt} = \frac{-1}{\sqrt{1-(t^{1/4})^2}} \times \frac{1}{4} t^{-3/4}$$

$$\boxed{\frac{d\theta}{dt} = \frac{-1}{\sqrt{1-t^{1/2}}} \cdot \frac{1}{4} t^{-3/4}}$$

$$\lim_{t \rightarrow 0} \theta = 1 - \cos \theta$$

$$\lim_{t \rightarrow 0} \theta = 1 - \sqrt{t}$$

$$\theta = 0 \quad t = 1$$

$$\theta = \pi/2 \quad t = 0$$

$$= \frac{1}{4} \int_1^0 \frac{1}{\sqrt{1-\frac{1}{2}(1-\sqrt{t})}} \times \frac{-1}{\sqrt{1-t}} \times t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \int_0^1 \frac{1}{\sqrt{1-\frac{1}{2}-\frac{\sqrt{t}}{2}}} \cdot \frac{1}{\sqrt{1-t}} t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \int_0^1 \frac{1}{\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{t}}} \times \frac{1}{\sqrt{1-t}} t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \int_0^1 \frac{1}{\sqrt{\frac{1}{4}(1+\sqrt{t})}} \times \frac{1}{\sqrt{1-t}} \times t^{-3/4} \cdot dt$$

$$= \frac{1}{4} \int_0^1 \frac{1}{\frac{1}{2} \sqrt{1+\sqrt{t}}} \times \frac{1}{\sqrt{1-t}} \times t^{-3/4} \cdot dt$$

$$= \frac{\sqrt{2}}{4} \int_0^1 \frac{1}{\sqrt{(1+t)(1-t)}} t^{-3/4} \cdot dt.$$

$$= \frac{\sqrt{2}}{4} \int_0^1 \frac{1}{\sqrt{1-(t^2)^2}} t^{-3/4} \cdot dt.$$

$$= \frac{\sqrt{2}}{4} \int_0^1 \frac{1}{\sqrt{1-t}} t^{-3/4} \cdot dt.$$

$$= \frac{\sqrt{2}}{4} \int_0^1 t^{-3/4} (1-t)^{-1/2} \cdot dt$$

$$= \frac{\sqrt{2}}{4} \times \pi \left(-\frac{3}{4} + 1, -\frac{1}{2} + 1 \right)$$

(L5,1) Differentiation under Integral Sign

$$I(d) = \int_a^b f(x, d) \cdot dx \quad \text{then} \quad \frac{dI}{dd} = \int_a^b \frac{\partial f}{\partial d} dx$$

Steps:-

- ① Parameter $\leftarrow$ Identity
- ② DOTS
- ③ Diff wto parameter.
- ④ Int
- ⑤ find 'c'

Q) PJ $\int_0^1 \frac{x^d - 1}{\log x} \cdot dx = \log(1+d) \quad d \geq 0$, Hence evaluate $\int_0^1 \frac{x^7 - 1}{\log x} \cdot dx$

$$\text{let } I(d) = \int_0^1 \frac{x^d - 1}{\log x} \cdot dx$$

by the rule of DOTS

$$\frac{dI}{dd} = \int_0^1 \frac{\partial}{\partial d} \left(\frac{x^d - 1}{\log x} \right) \cdot dx$$

$$= \int_0^1 \frac{1}{\log x} (x^d \log x) \cdot dx$$

$$x^d = x^{d \log x}$$

$$= \int_0^1 x^d \cdot dx$$

$$\frac{dI}{dx} = \left[\frac{x^{d+1}}{d+1} \right]_0^1$$

$$\frac{dI}{dx} = \frac{1}{d+1}$$

$$\int dI = \int \frac{1}{d+1} dx$$

$$I(d) = \log(d+1) + c$$

find c put $d=0$

$$I(0) = \log(0+1) + c \quad \text{--- (1)}$$

$$I(0) = \int_0^1 \frac{x^0 - 1}{\log x} \cdot dx$$

$$= \int_0^1 \frac{0}{\log x} \cdot dx$$

$$I(0) = 0$$

$$I(0) = \log 1 + c$$

$$|c = 0|$$

$$\text{from (1)} \quad I(d) = \log(d+1)$$

$$I = \log(7+1) = \log 8$$

Q) Show that $\int_0^{\infty} \frac{\log(1+ax^2)}{x^2} \cdot dx = \pi \sqrt{a} \quad (a > 0)$ evaluate $\int_0^a \frac{\log(1+x^2)}{x^2}$

$\cdot dx$

$$\text{let } I(a) = \int_0^{\infty} \frac{\log(1+au^2)}{u^2} \cdot du$$

by the rule of DILs

$$\frac{dI}{da} = \int_0^{\infty} \frac{\partial}{\partial a} \left(\frac{\log(1+au^2)}{u^2} \right) \cdot du$$

$$= \int_0^{\infty} \left[\frac{1}{u^2} \times \frac{1}{(1+au^2)} \cdot u^2 \right] \cdot du$$

$$= \int_0^{\infty} \left[\frac{1}{1+au^2} \right] \cdot du$$

$$\frac{dI}{da} = \int_0^{\infty} \frac{1}{1^2 + (\sqrt{a}u)^2} \cdot du$$

$$\boxed{\int \frac{1}{u^2 + a^2} \cdot du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right)}$$

$$= \left[\frac{1}{1} \tan^{-1} \left(\frac{\sqrt{a}u}{1} \right) \times \frac{1}{\sqrt{a}} \right]_0^{\infty}$$

$$= \frac{1}{\sqrt{a}} \left[\frac{\pi}{2} - 0 \right] = \frac{1}{\sqrt{a}} \frac{\pi}{2} = \frac{dI}{da}$$

$$\int dI = \int \frac{1}{\sqrt{a}} \frac{\pi}{2} \cdot da$$

$$I(a) = \frac{\pi}{2} \sqrt{a} + C$$

$$I(a) = \pi \sqrt{a} + c$$

put $a = 0$

$$I(0) = \pi \sqrt{0} + c$$

$$= 0$$

$$c = 0$$

from (I)

$$I(a) = \pi \sqrt{a}$$

⑥ Error functions

Error function = erf(x)

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} \cdot du.$$

Complementary error function erf(x)

$$\text{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-u^2} \cdot du.$$

properties of erf(x) :-

$$\text{erf}(\infty) = 1$$

$$\text{erf}(0) = 0$$

$$\text{erf}(x) + \text{erfc}(x) = 1$$

$$\text{erf}(-x) = -\text{erf}(x)$$

Differentiation of error function :-

$$\text{erf}(ax) = \frac{2}{\sqrt{\pi}} \int_0^{ax} e^{-u^2} \cdot du.$$

by the 2nd rule of DUIS

$$\frac{d}{dx} [\text{erf}(ax)] = \frac{2}{\sqrt{\pi}} \left[\int_0^{ax} \frac{d}{dx} e^{-u^2} \cdot du + \frac{d}{dx} (ax) e^{-a^2 x^2} - \frac{d}{du} (0) e^{-0^2} \right]$$

$$= \frac{2}{\sqrt{\pi}} \left[\int_0^{\infty} 0 \, du + e^{-a^2 u^2} - 0 \right]$$

Q) P.I. $\frac{d}{du} [\text{erf}(u)] = \frac{2}{\sqrt{\pi}} e^{-u^2}$ use it to show that $\frac{d}{du} [\text{erf}(au^n)] =$

$$\frac{2au^{n-1}}{\sqrt{\pi}} e^{-a^2 u^{2n}}$$

$$\text{erf}(u) = \frac{2}{\sqrt{\pi}} \int_0^u e^{-u^2} \cdot du.$$

by the 2nd rule of DUIS

$$\frac{d}{du} [\text{erf}(u)] = \frac{2}{\sqrt{\pi}} \left[\int_0^u \frac{d}{du} e^{-u^2} \cdot du + \frac{d}{du}(u) e^{-u^2} - \frac{d}{du}(0) e^{-0^2} \right]$$

$$= \frac{2}{\sqrt{\pi}} [0 + e^{-u^2} - 0]$$

$$= \frac{2}{\sqrt{\pi}} e^{-u^2}$$

$$\text{erf}(au^n) = \frac{2}{\sqrt{\pi}} \int_0^{au^n} e^{-u^2} \cdot du.$$

by the 2nd rule of DUIS

$$\frac{d}{du} [\text{erf}(au^n)] = \frac{2}{\sqrt{\pi}} \left[\int_0^{au^n} \frac{d}{du} e^{-u^2} \cdot du + \frac{d}{du}(au^n) \cdot e^{-a^2 u^{2n}} + \frac{d}{du}(0) e^{-0^2} \right]$$

$$= \frac{2}{\sqrt{\pi}} [0 + a n u^{n-1} \cdot e^{-a^2 u^{2n}} + 0]$$

$$= \frac{2}{\sqrt{\pi}} a n u^{n-1} \cdot e^{-a^2 u^{2n}}$$

Q) PT $\operatorname{erf}(-u) + \operatorname{erfc}(u) = 2$

$$\operatorname{erf}(u) + \operatorname{erfc}(u) = 1$$

$$\operatorname{erf}(-u) + \operatorname{erfc}(-u) = 1$$

$$\operatorname{erf}(-u) = -\operatorname{erf}(u)$$

$$-\operatorname{erf}(u) + \operatorname{erfc}(-u) = 1$$

$$\operatorname{erfc}(u) + \operatorname{erfc}(-u) = 1 + \operatorname{erf}(u) + \operatorname{erfc}(u)$$

$$\operatorname{erf}(u) + \operatorname{erf}(-u) = 1 + 1 = 2$$

Q) 55 $\int_0^{\infty} e^{-u^2 - 2bu} \cdot du = \frac{\sqrt{\pi}}{2} e^{b^2} [1 - \operatorname{erf}(b)]$

$$= \int_0^{\infty} e^{-u^2 - 2bu} \cdot du$$

$$= \int_0^{\infty} e^{-u^2 - 2bu - b^2 + b^2} \cdot du$$

$$= \int_0^{\infty} e^{-(u^2 + 2bu + b^2) + b^2} \cdot du$$

$$= \int_0^{\infty} e^{-(u+b)^2} e^{b^2}$$

$$= \int_0^{\infty} e^{-(u+b)^2} e^{b^2} \cdot du$$

$$= e^{b^2} \int_0^{\infty} e^{-(u+b)^2} \cdot du$$

$$\text{let } u+b = u$$

$$u = 0$$

$$u = b$$

$$u = \infty$$

$$u = \infty$$

$$\frac{du}{du} + 0 = 1$$

$$du = du$$

$$= e^{b^2} \int_0^{\infty} e^{-u^2} \cdot du$$

$$= e^{b^2} \frac{\sqrt{\pi}}{2} \left[\frac{2}{\sqrt{\pi}} \int_b^{\infty} e^{-u^2} \cdot du \right]$$

$$= e^{b^2} \frac{\sqrt{\pi}}{2} \times \operatorname{erfc}(b)$$

$$\operatorname{erf}(b) + \operatorname{erfc}(b) = 1$$

$$\operatorname{erfc}(b) = 1 - \operatorname{erf}(b)$$

$$= e^{b^2} \frac{\sqrt{\pi}}{2} [1 - \operatorname{erf}(b)]$$

⑦ Reduction formula

$$\int_0^{\pi/2} \sin^n x \, dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \dots \frac{2}{3} \text{ or } \frac{1}{2} \times p.$$

Even: 0, 2, 4, 6, 8

Stop here.

$$p = \pi/2 \dots \text{when } n = \text{even}.$$

$$p = 1 \dots \text{when } n = \text{odd}.$$

$$\int_0^{\pi/2} \sin^5 x \, dx = \frac{4}{5} \cdot \frac{2}{3} \times 1$$

$$\text{eg } \int_0^{\pi/2} \sin^4 x \, dx = \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{3\pi}{16}.$$

$$\int_0^{\pi/2} \sin^7 x \, dx = \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} = \frac{16}{35}$$

$$\int_0^{\pi/2} \sin^3 x \, dx.$$

$$\int_0^{\pi/2} \cos^n x \, dx =$$

$$\int_0^{\pi/2} \cos^8 x \, dx = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \times \frac{\pi}{2}$$

$$\int_0^{\pi/2} \cos^3 x \, dx = \frac{1}{2} \times 1$$

$$\int_0^{\pi/2} \sin^n u \cdot \cos^n u \cdot du = \frac{[(n-1)(n-3) \dots \dots 2 \text{ or } 1] [(n-1)(n-3) \dots \dots 2 \text{ or } 1]}{[(n+n) (n+n-2) (n+n-4) \dots \dots 2 \text{ or } 1]} \times P.$$

$$\text{eg } \int_0^{\pi/2} \sin^3 \theta \cos^4 \theta \cdot d\theta = \frac{[2][3 \cdot 1]}{7 \times 5 \times 3 \times 1}$$

$$P = \frac{\pi}{2} \quad n \& n \text{ even}$$

$$m=3, n=4 = \frac{2 \times 3}{7 \times 5 \times 3} = \frac{2}{35}$$

$$\int_0^{\pi} \sin^n x \cdot dx = 2 \int_0^{\pi/2} \sin^n u \cdot du \quad (\text{all values of } n)$$

$$\int_0^{\pi} \cos^n x \cdot dx = 2 \int_0^{\pi/2} \cos^n u \cdot du \quad (n \text{ is even integer})$$

$$= 0 \quad \dots \text{ Odd.}$$

$$\int_0^{2\pi} \sin^n x \cdot dx = 4 \int_0^{\pi/2} \sin^n u \cdot du \quad (n \text{ is even})$$

$$= 0 \quad n \text{ is odd.}$$

$$\int_0^{2\pi} \cos^n x \cdot dx = 4 \int_0^{\pi/2} \cos^n u \cdot du \quad (n \text{ is even})$$

$$= 0 \quad (n \text{ is odd})$$

$$\int_0^{\pi} \sin^n x \cdot \cos^m x \cdot dx = 2 \int_0^{\pi/2} \sin^n x \cdot \cos^m x \cdot dx \quad \text{if } n = \text{even} \quad m = e/o$$

$$= 0 \quad \text{if } n \text{ is odd} \quad m = e/o$$

$$\int_0^{2\pi} \sin^n x \cdot \cos^m x \cdot dx = 4 \int_0^{\pi/2} \sin^n x \cdot \cos^m x \cdot dx \quad \dots \dots \text{if } n, m = \text{even}$$

$$= 0 \quad \dots \dots \text{otherwise.}$$

$$\int_a^b u \cdot v \cdot dx = \left[u \int v \cdot dx \right]_a^b - \int_a^b \left[\frac{d}{dx} u \int v \cdot dx \right] \cdot dx.$$

L I A T E ✓

evaluate $\int_0^{\pi} \sin^7 x \cdot \cos^4 x \cdot dx$

$$= \int_0^{\pi} \sin^6 x \cdot \cos x \cdot dx = \int_0^{\pi} \sin^6(\pi - u) \cdot (-\cos u) \cdot du.$$

$$= \int_0^{\pi} (\pi - u) [\sin(\pi - u)]^6 \cos(\pi - u) \cdot du$$

$$\sin(\pi - u) = \sin u$$

$$\cos(\pi - u) = -\cos u.$$

$$= \int_0^{\pi} (\pi - u) [\sin u]^6 \cdot [-\cos u] \cdot du.$$

$$= \int_0^{\pi} (\pi - u) \sin^6 u \cdot \cos u \cdot du.$$

$$= \int_0^{\pi} (\sin^2 u \cdot \cos^4 u \cdot \pi) \cdot du - \pi \sin^2 u \cdot \cos^4 u \cdot du$$

$$= \int_0^{\pi} \pi \sin^2 u \cdot \cos^4 u \cdot du - \int_0^{\pi} \pi \sin^2 u \cdot \cos^4 u \cdot du.$$

$$I = \pi \int_0^{\pi} \sin^2 u \cos^4 u \cdot du - \int_0^{\pi} \pi \cdot \sin^2 u \cos^4 u \cdot du$$

$$I = \pi \int_0^{\pi} \sin^2 u \cdot \cos^4 u \cdot du - I.$$

$$= I + I = \pi \quad 2 \int_0^{\pi/2} \sin^2 u \cdot \cos^4 u \cdot du$$

$$\cancel{I} = \cancel{I} \int_0^{\pi/2} \sin^2 u \cdot \cos^4 u \cdot du.$$

$$I = \pi \left[\frac{(6 \times 4 \times 2)(3 \times 1)}{11 \times 9 \times 7 \times 5 \times 3 \times 1} \right]$$

$$I = \frac{\pi \times 6 \times 4 \times 2 \times 3}{11 \times 9 \times 7 \times 5 \times 3 \times 1} = \frac{16\pi}{155} //$$

⑧ Reduction formula 2

Q) Evaluate $\int_0^1 \frac{x^7}{\sqrt{1-x^2}} dx$

put $x = \sin \theta$

$$\frac{dx}{d\theta} = \cos \theta$$

$$dx = \cos \theta \cdot d\theta$$

if $\sqrt{1-x^2} \rightarrow$ then put $x = \sin \theta$

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$$

$$x = \sin \theta$$

$$0 = \sin \theta$$

$$\theta = \sin^{-1}(0)$$

$$\theta = 0$$

$$x = 0 \quad \theta = 0$$

$$x = 1 \quad \theta = \pi/2$$

$$I = \int_0^{\pi/2} \frac{(\sin \theta)^7}{\sqrt{1-(\sin \theta)^2}} \cos \theta \cdot d\theta$$

$$I = \int_0^{\pi/2} \frac{\sin^7 \theta}{\sqrt{\cos^2 \theta}} \times \cos \theta \cdot d\theta$$

$$I = \int_0^{\pi/2} \frac{\sin^7 \theta}{\cos \theta} \times \cancel{\cos \theta} \cdot d\theta$$

$$I = \int_0^{\pi/2} \sin^7 \theta \cdot d\theta$$

$$= \frac{6^2}{7} \frac{4}{5} \frac{2}{3} = 16/35$$

$$Q_2) \int_0^1 x^5 \sin^{-1} x \cdot dx = I$$

$\downarrow$
 A
 $\odot$
 u

$\downarrow$
 I
 $\odot$
 v

Use LIATE Rule

$$\int_a^b u \cdot v \cdot dx = \left[u \int v \right]_a^b - \int_a^b \left[\frac{d}{dx} u \int v \cdot dx \right] \cdot dx$$

$$= \left[\sin^{-1} x \int x^5 \cdot dx \right]_0^1 - \int_0^1 \left[\frac{d}{dx} \sin^{-1} x \int x^5 dx \right] \cdot dx$$

$$= \left[\sin^{-1} x \cdot \frac{x^6}{6} \right]_0^1 - \int_0^1 \left[\frac{1}{\sqrt{1-x^2}} \cdot \frac{x^6}{6} \right] \cdot dx$$

$$= \left[\left(\frac{\pi}{2} \cdot \frac{1}{6} \right) \right] - \frac{1}{6} \int_0^1 \left[\frac{x^6}{\sqrt{1-x^2}} \cdot dx \right]$$

$$I = \frac{\pi}{12} - \frac{1}{6} I_1 \quad - (1)$$

$$I_1 = \int_0^1 \frac{x^6}{\sqrt{1-x^2}} \cdot dx$$

put $x = \sin \theta$

$$\frac{dx}{d\theta} = \cos \theta$$

$$\boxed{dx = \cos \theta \cdot d\theta}$$

$$x = 0 \quad \theta = 0$$

$$x = 1 \quad \theta = \pi/2$$

$$I_1 = \int_0^{\pi/4} \frac{(\sin \theta)^6}{\sqrt{1 - (\sin \theta)^2}} \times \cos \theta \cdot d\theta$$

$$I_1 = \int_0^{\pi/4} \frac{\sin^6 \theta}{\sqrt{\cos^2 \theta}} \times \cos \theta \cdot d\theta$$

$$I_1 = \int_0^{\pi/4} \frac{\sin^6 \theta}{\cancel{\cos \theta}} \cdot \cancel{\cos \theta} \cdot d\theta$$

$$I_1 = \int_0^{\pi/4} \sin^6 \theta \cdot d\theta = \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \boxed{\frac{5\pi}{32}}$$

from (I)

$$I = \frac{\pi}{12} - \frac{1}{6} \frac{5 \times 3 \times \pi}{6 \times 4 \times 2 \times 2}$$

$$I = \frac{\pi}{12} \left[1 - \frac{15}{48} \right]$$

$$I = \frac{\pi}{12} \left(\frac{48 - 15}{48} \right) = \frac{\pi}{12} \times \frac{33}{48} = \frac{33\pi}{512 \times 16} = \frac{11\pi}{192}$$

Q) If $I_n = \int_0^{\pi/4} \frac{\sin(2n-1)u}{\sin u} du$ then prove that $n(I_{n+1} - I_n) = \sin \frac{n\pi}{2}$
 & then find I_5

$$I_n = \int_0^{\pi/4} \frac{\sin(2n-1)u}{\sin u} \cdot du$$

$$I_{n+1} = \int_0^{\pi/4} \frac{\sin(2(n+1)-1)u}{\sin u} \cdot du$$

$$= \int_0^{\pi/4} \frac{\sin(2n+1)x}{\sin x} x dx = \int_0^{\pi/4} \frac{\sin(Lx+1)x}{\sin x} x dx.$$

$$I_{n+1} - I_n = \int_0^{\pi/4} \frac{\sin(2n+1)x}{\sin x} x dx - \int_0^{\pi/4} \frac{\sin(Lx+1)x}{\sin x} x dx$$

$$= \int_0^{\pi/4} \left[\frac{\sin(Ln+1)x}{\sin x} - \frac{\sin(Ln-1)x}{\sin x} \right] x dx.$$

$$= \int_0^{\pi/4} \frac{(\sin(2n+1)x - \sin(2n-1)x)}{\sin x} x dx$$

$$\left[\sin c - \sin 0 = 2 \cos \left(\frac{c+0}{2} \right) \sin \left(\frac{c-0}{2} \right) \right]$$

$$I_{n+1} - I_n = \int_0^{\pi/4} \frac{2 \cdot \cos 2nx \cdot \cancel{\sin x}}{\cancel{\sin x}} x dx$$

$$= 2 \int_0^{\pi/4} \cos 2nx \cdot x dx$$

$$I_{n+1} - I_n = \left[2 \frac{\sin 2nx}{2n} \right]_0^{\pi/4}$$

$$n(I_{n+1} - I_n) = \sin \frac{n\pi}{2}$$

$$n(I_{n+1} - I_n) = \sin \frac{n\pi}{2}$$

$$I_3 - (I_2 - I_1) = 1$$

$$I_L - I_r = 1$$

$$2(I_L - I_L) = 0$$

$$I_3 - I_L = 0$$

$$I_3 = I_2$$

$$I_1 = \int_0^{1/4} \frac{\cancel{\lambda_2} \cdot \cancel{\pi}}{\cancel{\lambda_2} \cdot \cancel{\pi}} \cdot du$$

$$= (u) \Big|_0^{1/4}$$

$$= \frac{\pi}{4} - 0 = \pi/4.$$

$$I_1 = \pi/4$$

$$I_3 = I_2$$

$$I_3 = I_2 = 1 + I_1 = 1 + \pi/4$$

⑨ Reduction formula 3

Q) If $u_n = \int_0^{\pi/4} \sin^{2n} x \cdot dx$ P.T $u_n = \left(1 - \frac{1}{2^n}\right) u_{n-1} - \frac{1}{n \cdot 2^{n+1}}$

$$u_n = \int_0^{\pi/4} \sin^{2n} x \cdot dx.$$

$$u_n = \int_0^{\pi/4} \sin^{2n-1} x \cdot \sin x \cdot dx. \quad \text{--- To apply } u \cdot v \text{ rule}$$

$$u_n = \left[\sin^{2n-1} x \int \sin x \cdot dx \right]_0^{\pi/4} - \int_0^{\pi/4} \left(\frac{d}{dx} \sin^{2n-1} x \int \sin x \cdot dx \right) \cdot dx.$$

$$u_n = \left[\sin^{2n-1} x (-\cos x) \right]_0^{\pi/4} - \int_0^{\pi/4} \left[(2n-1) (\sin x)^{2n-2} \cdot (\cos x) \cdot (-\cos x) \right] \cdot dx$$

$$u_n = \left[-\left(\frac{1}{\sqrt{2}}\right)^{2n-1} \left(\frac{1}{\sqrt{2}}\right) \right] - \int_0^{\pi/4} \left[(2n-1) \sin^{2n-2} x \cos^2 x \right] \cdot dx.$$

$$u_n = \left[-\left(\frac{1}{\sqrt{2}}\right)^{2n} \right] + \int_0^{\pi/4} \left[(2n-1) \sin^{2n-2} x (1 - \sin^2 x) \right] \cdot dx.$$

$$u_n = \left[-\frac{1}{\left(\frac{1}{\sqrt{2}}\right)^{2n}} \right] + \int_0^{\pi/4} \left[(2n-1) \sin^{2n-2} x - (2n-1) \sin^{2n-2} x \sin^2 x \right] \cdot dx.$$

$$u_n = \frac{-1}{2^n} + (2n-1) \int_0^{\pi/4} \sin^{2n-2} x \cdot dx - (2n-1) \int_0^{\pi/4} \sin^{2n} x \cdot dx.$$

$$u_n = \frac{-1}{2^n} + (2n-1) u_{n-1} - (2n-1) u_n$$

$$u_n = \left(1 - \frac{1}{2^n}\right) u_{n-1} - \frac{1}{n \cdot 2^{n+1}}$$

$$u_n + (2n-1) u_n = \frac{-1}{2^n} + (2n-1) u_{n-1}$$

$$u_n (\cancel{1} + 2n - 1) = \frac{-1}{2^n} + (2n-1) u_{n-1}$$

$$u_n (2n) = \frac{-1}{2^n} + (2n-1) u_{n-1}$$

$$u_n = \frac{-1}{2^n \cdot 2n} + \frac{(2n-1)}{2n} u_{n-1}$$

$$u_n = \frac{-1}{n \cdot 2^{n+1}} + \left(1 - \frac{1}{2n}\right) u_{n-1}$$

Q2) If $I_n = \int_0^{\infty} e^{-x} \lambda \sin^n x \cdot dx$ obtain the relation bet I_n & I_{n-2} & hence

find I_4

$$I_n = \int_0^{\infty} \underbrace{e^{-x}}_E \underbrace{\lambda \sin^n x}_{T} dx$$

(V) (W)

$$I_n = \left[\lambda \sin^n x \int e^{-x} \right]_0^{\infty} - \int_0^{\infty} \left[\frac{d}{dx} \lambda \sin^n x \right] \left[e^{-x} \right] \cdot dx$$

$$I_n = \left[\lambda \sin^n x \cdot \frac{e^{-x}}{-1} \right]_0^{\infty} - \int_0^{\infty} \left[n \cdot (\lambda \sin x)^{n-1} \cos x \cdot \frac{e^{-x}}{-1} \right] dx$$

$$I_n = 0 + n \int_0^{\infty} \left[\frac{\sin^{n-1} x \cdot \cos x}{x} \cdot \frac{e^{-x}}{x} \right] \cdot dx.$$

$$I_n = n \left\{ \left[\sin^{n-1} x \cdot \cos x \int e^{-x} \right]_0^{\infty} - \int_0^{\infty} \frac{d}{dx} (\sin^{n-1} x \cdot \cos x) \cdot \left[e^{-x} dx \right] \cdot dx \right\}$$

$$I_n = n \left\{ \left[\sin^{n-1} x \cdot \cos x \frac{e^{-x}}{-1} \right]_0^{\infty} - \int_0^{\infty} \left[\sin^{n-1} x (-\sin x) + \cos x (n-1) (\sin x)^{n-2} \cdot \cos x \cdot \frac{e^{-x}}{-1} \right] \cdot dx \right\}$$

$$I_n = n \left\{ 0 - \int_0^{\infty} \left[(\sin^{n-1} x \cdot \sin x + (n-1) \sin^{n-2} x \cdot \cos^2 x) e^{-x} \right] \cdot dx \right\}$$

$$I_n = n \int_0^{\infty} \left[-\sin^n x + (n-1) \sin^{n-2} x (1 - \sin^2 x) \right] e^{-x} \cdot dx.$$

$$I_n = n \int_0^{\infty} \left[(-\sin^n x + (n-1) \sin^{n-2} x - (n-1) \sin^n x) e^{-x} \right] \cdot dx.$$

$$I_n = n \left[- \int_0^{\infty} e^{-x} \sin^n x \cdot dx + (n-1) \int_0^{\infty} e^{-x} \sin^{n-2} x \cdot dx - (n-1) \int_0^{\infty} e^{-x} \sin^n x \cdot dx \right]$$

$$I_n = n \left[-I_n + (n-1) I_{n-2} - (n-1) I_n \right]$$

$$I_n = n \left[-\cancel{I_n} + (n-1) I_{n-2} - n I_n + \cancel{I_n} \right]$$

$$I_n = n(n-1) I_{n-2} - n^2 I_n$$

$$I_n + n^2 I_n = n(n-1) I_{n-2}$$

$$I_n(1+n^2) = n(n-1) I_{n-2}$$

$$I_n(1+n^2) = n(n-1) I_{n-2}$$

$$I_n = \frac{n(n-1)}{(1+n^2)} I_{n-2}$$

$$I_4 = \frac{4(4-1)}{1+16} I_2 = \frac{12}{17} I_2$$

$$I_2 = \frac{2(1)}{5} I_0$$

$$I_4 = \frac{12}{17} \times \frac{2}{5} I_0$$

$$I_4 = \frac{24}{85} (i)$$

$$I_n = \int_0^{\infty} e^{-x} x^n dx$$

$$I_0 = \int_0^{\infty} e^{-x} \cdot dx$$

$$= \left(\frac{e^{-x}}{-1} \right)_0^{\infty}$$

$$= \{1\}$$

④ Double Integrals

Or Multiple Integrals

① Evaluation of Double Integrals

Formulas :- refer the calculus book by T.A. maron.

$$\int a^n dx = \frac{a^n}{\log a} \quad \int \sin x \cdot dx = -\cos x$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left(\frac{x-a}{x+a} \right) \quad \int \frac{1}{a^2 - x^2} = \frac{1}{2a} \log \left(\frac{a+x}{a-x} \right)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) \quad \int \frac{1}{\sqrt{x^2 + a^2}} = \log |x + \sqrt{x^2 + a^2}|$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}|$$

$$\int \frac{f'(x)}{f(x)} \cdot dx = \log |f(x)|$$

$$e.g. \int \frac{2x+3}{x^2+3x+5} dx = \log(x^2+3x+5)$$

$$\int \frac{f'(x)}{\sqrt{f(x)}} dx = 2\sqrt{f(x)}$$

$$\int \cos(3x+2) dx.$$

$$\int x^5 dx$$

$$I = \frac{\sin(3x+1)}{3}$$

$$= \frac{x^6}{6}$$

$$\int_{x=}$$

$$\int_{y=}$$

$$dy$$

$$I = \int_{y=0}^1 \int_{x=0}^2 xy \cdot dx \cdot dy$$

$$I = \int_0^4 \int_0^x xy \cdot dx \cdot dy$$

$$I = \int_0^4 \int_0^{\frac{y}{2}} xy \cdot dx \cdot dy$$

$$I = \int_0^4 \left[\int_{y=0}^{\frac{y}{2}} xy \cdot dy \right] dx$$

$$I = \int_0^4 x \left[\frac{y^2}{2} \right]_0^{\frac{y}{2}} dx$$

$$I = \int_0^4 x \left[\frac{x^2}{2} \right] dx$$

$$= \int_0^4 \frac{x^3}{2} \cdot dx.$$

$$I = \int_0^5 \int_0^4 x y \cdot dx \cdot dy$$

$$I = \int_0^5 \left[\int_{x=0}^{x=y} x y \cdot dx \right] dy$$

$$\textcircled{1} \int_0^1 \int_0^x x y \cdot dy \cdot dx$$

$$I = \int_0^1 \left[\int_{y=0}^{y=x} x y \cdot dy \right] dx.$$

$$I = \int_0^1 \left[\int_0^x x y \cdot dy \right] dx$$

$$I = \int_0^1 \left[x \cdot \frac{y^2}{2} \right]_0^x dx.$$

$$I = \int_0^1 \left[x \cdot \frac{x^2}{2} - x \cdot \frac{0^2}{2} \right] dx.$$

$$= \int_0^1 \left[\frac{x^3}{2} \right] dx = \frac{1}{2} \int_0^1 x^3 dx.$$

$$= \frac{1}{2} \left[\frac{x^4}{4} \right]_0^1 = \frac{1}{2} \left[\frac{1}{4} - \frac{0}{4} \right] = \frac{1}{8}.$$

$$\textcircled{1} \int_0^2 \int_0^{\sqrt{2u}} xy \cdot dy \cdot du$$

$$I = \int_0^2 \left[\int_0^{\sqrt{2u}} xy \cdot dy \right] du$$

$$I = \int_0^2 \left[x \cdot \frac{y^2}{2} \right]_0^{\sqrt{2u}} du$$

$$I = \int_0^2 \left[x \cdot \frac{(\sqrt{2u})^2}{2} - x \cdot \frac{0^2}{2} \right] \cdot du$$

$$I = \int_0^2 \left[\frac{2x^2}{2} \right] du$$

$$I = \int_0^2 x^2 \cdot du$$

$$I = \left[\frac{x^3}{3} \right]_0^2 = \left(\frac{2^3}{3} - \frac{0^3}{3} \right)$$

$$I = \frac{8}{3}$$

$$\textcircled{2} \int_0^1 \int_0^u (u^2 + y^2) u \cdot dy \cdot du$$

$$I = \int_0^1 \left[\int_0^u (u^2 + y^2) \cdot dy \right] du$$

$$I = \int_0^1 \left[x^3 y + \frac{xy^3}{3} \right]_0^1 dx.$$

$$= \int_0^1 \left[x^3 \cdot 1 + \frac{x \cdot 1^3}{3} \right] dx$$

$$I = \int_0^1 \left[x^3 + \frac{x}{3} \right] dx.$$

$$I = \left[\frac{x^4}{4} + \frac{1}{3} \frac{x^2}{2} \right]_0^1$$

$$I = \left[\frac{1}{4} + \frac{1}{3} \cdot \frac{1}{2} \right] = \frac{1}{4} + \frac{1}{6} = \frac{4}{12}$$

(4) find $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dy dx.$

$$I = \int_0^1 \left[\int_0^{\sqrt{1+x^2}} \frac{1}{\underbrace{y^2 + (1+x^2)}_{x^2 + 0^2}} dy \right] dx. \quad \left(\frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) \right).$$

$$I = \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} \left(\frac{y}{\sqrt{1+x^2}} \right) \right]_0^{\sqrt{1+x^2}} dx$$

$$I = \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1} \left(\frac{\cancel{1+x^2}}{\sqrt{1+x^2}} \right) \right] dx.$$

$$I = \int_0^1 \left[\frac{1}{\sqrt{1+x^2}} \tan^{-1}(1) \right] dx.$$

$$I = \int_0^1 \left[\frac{1}{1+u^2} \cdot \frac{\pi}{4} \right] \cdot du.$$

$$I = \frac{\pi}{4} \int_0^1 \frac{1}{1+u^2} du.$$

$$\int \frac{1}{1+u^2} du = \log |u + \sqrt{1+u^2}|$$

$$I = \frac{\pi}{4} \int_0^1 \frac{1}{1+u^2} \cdot du.$$

$$I = \frac{\pi}{4} \left[\log(u + \sqrt{1+u^2}) \right]_0^1$$

$$I = \frac{\pi}{4} \left[\log(1 + \sqrt{2}) \right]$$

$$\textcircled{5} \quad I = \int_0^{\sqrt{3}} \int_0^{\sqrt{x^2+9}} \frac{u}{u^2+x^2+9} du \cdot dx$$

$$I = \int_0^{\sqrt{3}} \left[\int_0^{\sqrt{x^2+9}} \frac{u}{u^2+x^2+9} du \right] dx. \quad \left(\frac{1}{x^2+a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) \right)$$

$$I = \int_0^{\sqrt{3}} x \cdot \left[\frac{1}{\sqrt{x^2+9}} \tan^{-1} \left(\frac{u}{\sqrt{x^2+9}} \right) \right]_0^{\sqrt{x^2+9}} dx.$$

$$I = \int_0^{\sqrt{3}} x \left[\frac{1}{\sqrt{x^2+9}} \cdot \frac{\pi}{4} \right] \cdot dx.$$

$$I = \frac{1}{4} \frac{1}{2} \int_0^{2\sqrt{3}} \frac{2u}{\sqrt{u^2 + a^2}} \cdot du.$$

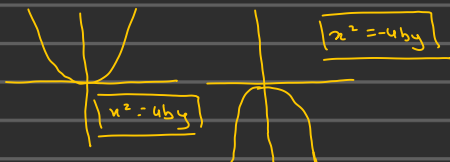
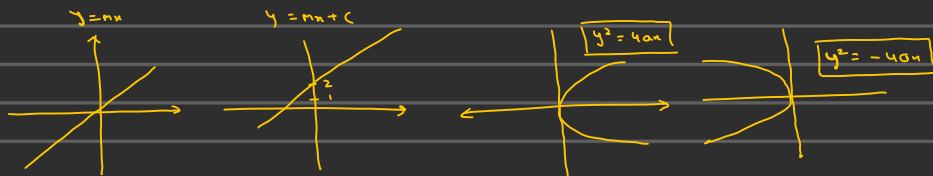
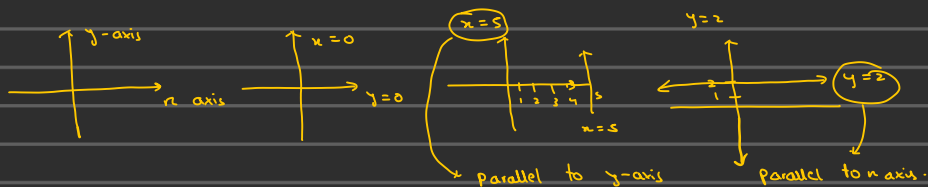
$$\int \frac{f'(u)}{\sqrt{f(u)}} = 2\sqrt{f(u)}$$

$$I = \frac{\pi}{8} \left[2\sqrt{u^2 + a^2} \right]_0^{2\sqrt{3}} = \frac{\pi}{8} \left[2\sqrt{3a^2 + a^2} - 2\sqrt{a^2} \right]$$

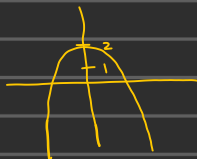
$$= \frac{\pi}{8} \left[2\sqrt{4a^2} - 2a \right] = \frac{\pi}{8} [2 \times 2a - 2a]$$

$$= \frac{\pi}{8} [2a] = \frac{\pi}{4} a.$$

② Change the order of Double Integral



$$x^2 = -y + 2$$



$$x^2 = -y - 2$$



$$x^2 + y^2 = 1$$

$$y = \sqrt{1-x^2}$$

$$x=5, y=\sqrt{1-x^2}$$

$$\int_{x=0}^5 \int_{y=0}^{\sqrt{1-x^2}} dy \, dx.$$

$$\int_{y=0}^1 \int_{x=0}^{\sqrt{1-y^2}} dx \, dy$$

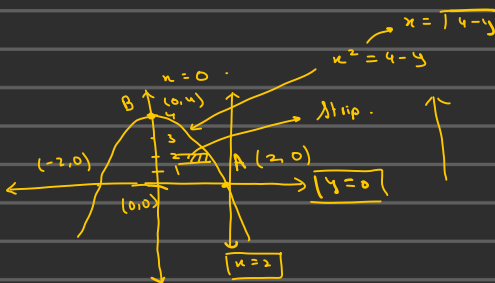
Q.) change the order of Integration & evaluate integral

$$\int_{x=0}^2 \int_{y=0}^{4-x^2} \frac{x \cdot e^{2y}}{4-y} dy \, dx$$

$$\boxed{x=0} \quad \boxed{y=0} \quad \checkmark$$

$$\boxed{x=2} \quad \boxed{y=4-x^2} \quad \checkmark$$

$$\boxed{x^2 = -y+4} \quad \checkmark$$



find coordinate of A =

$$x^2 = -y+4$$

$$y=0$$

to solve $x^2 = -y+4$ & $y=0$

$$x^2 = -0+4$$

$$\boxed{x = \pm 2}$$

= change the order of Integration

$$= \int_{y=0}^4 \int_{x=0}^{\sqrt{4-y}} \frac{x \cdot e^{2y}}{4-y} dy \, dx.$$

in x you'd need to go $\longrightarrow$

$$= \int_0^4 \left[\int_0^{\sqrt{4-y}} \frac{x \cdot e^{2y}}{4-y} dx \right] dy$$

$$= \int_0^4 \left[\frac{e^{2y}}{4-y} \cdot \frac{x^2}{2} \right]_0^{\sqrt{4-y}} dy$$

$$= \int_0^4 \left[\frac{e^{2y}}{4-y} \cdot \frac{(\sqrt{4-y})^2}{2} \right] dy$$

$$= \int_0^4 \left[\frac{e^{2y}}{\cancel{4-y}} \cdot \frac{\cancel{(4-y)}}{2} \right] dy$$

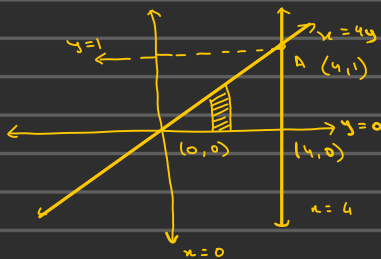
$$= \frac{1}{2} \int_0^4 e^{2y} dy = \frac{1}{2} \left[\frac{e^{2y}}{2} \right]_0^4$$

$$= \frac{1}{4} [e^{2y}]_0^4 = \frac{1}{4} [e^{2 \cdot 4} - e^{2 \cdot 0}] = \frac{1}{4} (e^8 - 1)$$

$$Q2) \int_{y=0}^{y=1} \int_{x=0}^{x=4y} e^{x^2} dx dy$$

$$y=0 \quad \boxed{x=4y}$$

$$y=1 \quad x=4$$

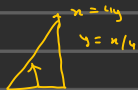


find A = (4,1)

$$x=4y \quad x=4y$$

$$\boxed{y=1}$$

$$\int_{x=0}^{x=4} \left[\int_{y=0}^{y=x/4} e^{x^2} dy \right] dx$$



$$= \int_0^4 \left[e^{x^2} y \right]_0^{x/4} dx = \int_0^4 \left[e^{x^2} \frac{x}{4} \right] dx$$

$$= \frac{1}{4} \int_0^4 [e^{x^2} x] dx$$

Put $u^2 = t$

$$2u = dt/du$$

$$u du = dt/2$$

$$= \frac{1}{4} \int_0^{16} e^t \frac{dt}{2}$$

$$= \frac{1}{8} \int_0^{16} e^t \cdot dt = \frac{1}{8} [e^t]_0^{16}$$

$$= \frac{1}{8} [e^{16} - e^0] = \frac{1}{8} [e^{16}]$$

Q₃) $\int_{u=0}^{u=1} \int_{y=0}^{y=\sqrt{1-u^2}} \frac{e^y}{(e^y+1)\sqrt{1-u^2-y^2}} dy du$

$$u=0$$

$$y=0$$

$$u=1$$

$$y = \sqrt{1-u^2}$$

$$y' = 1-u^2$$

$$\boxed{u^2 + y^2 = 1}$$

$$= \int_0^1 \left[\int_0^{\sqrt{1-u^2}} \frac{e^y}{(e^y+1)\sqrt{1-u^2-y^2}} dy \right] du$$

$$u^2 = 1-y^2$$

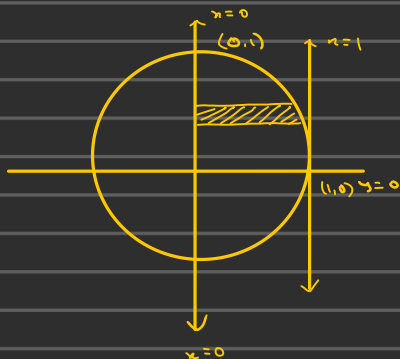
$$u = \sqrt{1-y^2}$$

$$= \int_0^1 \left[\int_0^{\sqrt{1-y^2}} \frac{e^y}{(e^y+1)\sqrt{1-y^2-u^2}} du \right] dy$$

$$\int \frac{1}{\sqrt{a^2-u^2}} du = \sin^{-1}\left(\frac{u}{a}\right)$$

$$= \int_0^1 \left[\frac{e^y}{e^y+1} \sin^{-1}\left(\frac{u}{\sqrt{1-y^2}}\right) \right]_0^{\sqrt{1-y^2}} dy$$

$u=0$	$t=u^2$	$t=0$
$u=4$	$t=u^2$	$t=16$



$$= \int_0^1 \left(\frac{e^y}{e^y + 1} \cdot \frac{\pi}{2} \right) dy = \frac{\pi}{2} \int_0^1 \frac{e^y}{e^y + 1} dy$$

$$= \frac{\pi}{2} \left[\log(e^y + 1) \right]_0^1$$

$$= \frac{\pi}{2} \left[\log(e^1 + 1) - \log(e^0 + 1) \right]$$

$$= \frac{\pi}{2} \left[\log(e + 1) - \log 2 \right]$$

$$= \frac{\pi}{2} \log\left(\frac{e+1}{2}\right)$$

$$\int \frac{f'(x)}{f(x)} dx = \log f(x)$$

③ Change the order of Int.

Q) change the order of Integration & evaluate

$$I = \int_{x=0}^{x=5} \int_{y=2-x}^{y=2+x} dz \, dx$$

$$x = 0$$

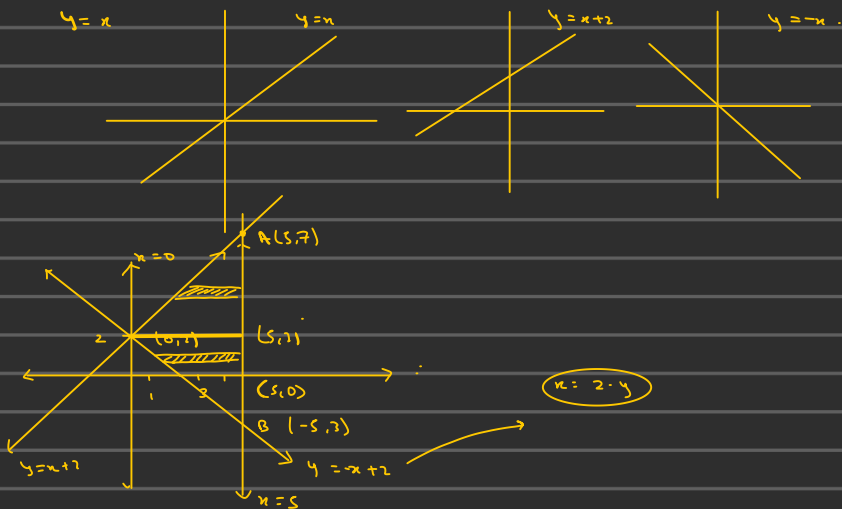
$$y = 2+x$$

$$y = x+2$$

$$x = 5$$

$$y = 2-x$$

$$y = -x+2$$



$$x = 2-y$$

find $A = (?)$

$$x = 5 \quad \& \quad y = x+2$$

$$y = 5+2$$

$$y = 7$$

$B \equiv (?)$

$$x = 5 \quad \& \quad y = -x+2$$

$$y = -5+2$$

$$y = -3$$

$$(5, -7)$$

$$= \int_{-3}^7 \left[\int_{2-y}^5 dz \right] dy + \int_2^7 \left[\int_{5-2}^5 dz \right] dy$$

$$= \int_{-3}^2 \left[\frac{x^5}{2-y} \right]_{2-y} dy + \int_2^7 \left[\frac{x^5}{y-2} \right]_{y-2} dy$$

$$= \int_{-3}^2 [5 - (2-y)] \cdot dy + \int_2^7 [5 - y - 2] \cdot dy$$

$$= \int_{-3}^2 [3+y] \cdot dy + \int_2^7 [7-y] dy$$

$$= \left[3y + \frac{y^2}{2} \right]_{-3}^2 + \left[7y - \frac{y^2}{2} \right]_2^7$$

$$= \left[6 + \frac{4}{2} - \left(-9 + \frac{9}{2} \right) \right] + \left[49 - \frac{49}{2} - \left(14 - \frac{4}{2} \right) \right]$$

$$= \left[6 + 2 + 9 - \frac{9}{2} \right] + \left[49 - \frac{49}{2} - 14 + 2 \right]$$

$$= \left[17 - \frac{9}{2} \right] - \left[37 - \frac{49}{2} \right] = \left[\frac{34-9}{2} \right] + \left[\frac{74-49}{2} \right]$$

$$= \frac{25}{2} + \frac{25}{2} = \underline{\underline{25}}$$

$$Q1) \int_{x=0}^{x=1} \int_{y=x}^{y=2-x} \frac{x}{y} dy \cdot dx$$

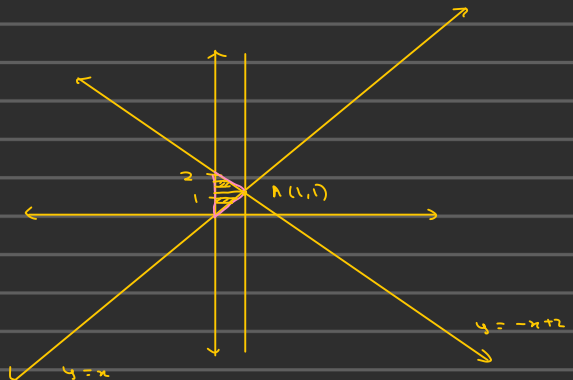
$$x=0 \quad y=x$$

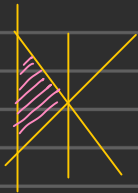
$$x=1 \quad y=2-x$$

$$y = -x+2$$

To solve $y=x$ & $y = -x+2$

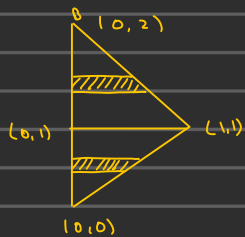
$$\boxed{y=1}$$





$$y = -x + 2 \quad \& \quad x = 0$$

$$\boxed{y = 2} \quad ; \quad \boxed{z = 0}$$



$$= \int_0^1 \int_0^y \frac{x}{y} \, dx \, dy + \int_1^2 \int_0^{2-y} \frac{x}{y} \, dx \, dy$$

④ Change of Integral from Cartesian to polar

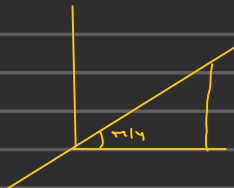
Cartesian (x, y)
polar (r, θ)

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x = y$$

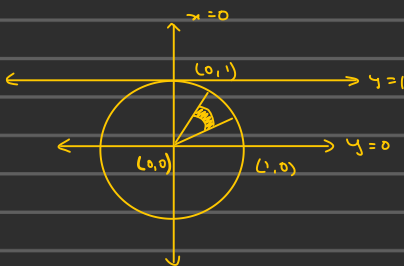
$$dx dy = r dr d\theta$$



Q1) change to polar coordinate & evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$

$$\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$$

$$dx dy = r dr d\theta$$



$$y=0 \quad x=0$$

$$y=1 \quad x = \sqrt{1-y^2} \quad x^2 = 1-y^2$$

$$\boxed{x^2 + y^2 = 1^2}$$

Strip always $\parallel$ to y-axis.

$$r^2 (\cos^2 \theta + \sin^2 \theta) = 1$$

$$r^2 (\cos^2 \theta + \sin^2 \theta) = 1$$

$$r^2 = 1 \quad \boxed{r=1}$$

$$r = 0 \text{ to } r=1 \quad \theta=0 \text{ to } \theta=\pi/2$$

$$= \int_0^{\pi/2} \int_0^1 \underbrace{r^2 \cdot r}_{r^3} dr d\theta$$

$$x^2 + y^2 = (r \cos \theta)^2 + (r \sin \theta)^2$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$x^2 + y^2 = 1^2$$

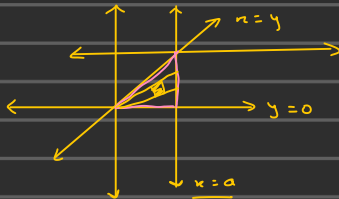
$$= \int_0^{\pi/2} \left[\frac{r^4}{4} \right]_0^1 \cdot d\theta$$

$$= \int_0^{\pi/2} \left[\frac{1}{4} \right] d\theta = \frac{1}{4} \int_0^{\pi/2} d\theta$$

$$= \frac{1}{4} [\theta]_0^{\pi/2} = \frac{1}{4} \left[\frac{\pi}{2} \right] = \boxed{\frac{\pi}{8}}$$

Q2) $\int_{y=0}^{y=a} \int_{x=y}^{x=a} x \cdot dx \cdot dy$

$y=0 \quad y=a$
 $x=y \quad x=a$

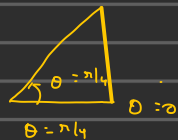


$r=0 \quad r=?$

$x=a$

$r \cos \theta = a$

$$r = \frac{a}{\cos \theta}$$



$\theta = 0$

$\theta = ?$

$x=y$

$\tan^{-1}(1) = \theta$

$r \cos \theta = 1/\sin \theta$

$1 = \frac{\sin \theta}{\cos \theta}$

$$\theta = \pi/4$$

$1 = \tan \theta$

$$= \int_0^{\pi/4} \int_0^{a/\cos \theta} x \cos \theta \cdot r \cdot dr \cdot d\theta$$

$$= \int_0^{\pi/4} \int_0^{a/\cos \theta} \left[r^2 \cos \theta \right] dr \cdot d\theta = \int_0^{\pi/4} \left[\cos \theta \cdot \frac{r^3}{3} \right]_0^{a/\cos \theta} d\theta$$

$$= \int_0^{\pi/4} \left[\cos \theta \times \frac{a^2}{3 \cos^3 \theta} \right] \cdot d\theta$$

$$= \int_0^{\pi/4} \left[\cancel{\cos \theta} \times \frac{a^2}{3 \cos^2 \theta} \right] \cdot d\theta = \frac{a^2}{3} \int_0^{\pi/4} \sec^2 \theta \cdot d\theta$$

$$= \frac{a^2}{3} [\tan \theta]_0^{\pi/4} = \frac{a^2}{3} \left[\tan \frac{\pi}{4} \right] = \frac{a^2}{3}$$

$$Q_{11}) \int_0^1 \int_{\pi}^{\sqrt{2\pi-x^2}} (x^2 + y^2) dy \cdot dx$$

$$x=0$$

$$y=x$$

$$x=1$$

$$y = \sqrt{2\pi - x^2}$$

$$y^2 = 2\pi - x^2$$

$$x^2 - 2x + y^2 = 0$$

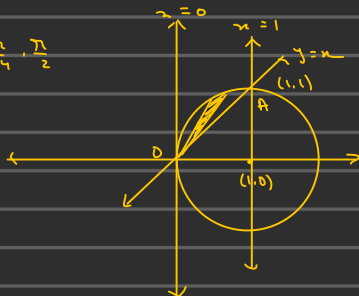
$$x^2 - 2(x)(1) + 1^2 - 1^2 + y^2 = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

$$(x-1)^2 + y^2 = 1$$

$$C = (1,0)$$

$$\theta = \frac{\pi}{4}, \frac{\pi}{2}$$



$$x^2 - 2x + y^2 = 0$$

$$r=0$$

$$y^2 = 2\pi - x^2$$

$$x^2 - 2x + 1^2 = 0$$

$$y^2 + x^2 = 2\pi$$

$$x^2 - 2x = 0$$

$$r^2 \sin^2 \theta + r^2 \cos^2 \theta = 2r \cos \theta$$

$$2x(x-1) = 0$$

$$r^2 (\sin^2 \theta + \cos^2 \theta) = 2r \cos \theta$$

$$\boxed{x=0} \quad \boxed{x=1}$$

$$r^2 (1) = 2r \cos \theta$$

$$\boxed{r = 2 \cos \theta}$$

$$\int_{\pi/4}^{\pi/2} \int_0^{2 \cos \theta} r^2 \cdot r \cdot dr \cdot d\theta$$

$$= \int_{\pi/4}^{\pi/2} \left[\frac{r^4}{4} \right]_0^{2 \cos \theta} \cdot d\theta$$

$$= \int_{\pi/4}^{\pi/2} \left[\frac{16(\cos \theta)^4}{4} \right] \cdot d\theta.$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$= 4 \int_{\pi/4}^{\pi/2} [(\cos^2 \theta)^2 \cdot d\theta] = 4 \int_{\pi/4}^{\pi/2} \left[\left(\frac{1 + \cos 2\theta}{2} \right)^2 \right] \cdot d\theta$$

$$= 4 \int_{\pi/4}^{\pi/2} \left[\frac{(1 + \cos 2\theta)^2}{4} \right] d\theta$$

$$= \int_{\pi/4}^{\pi/2} [1 + 2\cos 2\theta + (\cos 2\theta)^2] \cdot d\theta$$

$$= \int_{\pi/4}^{\pi/2} \left[1 + 2\cos 2\theta + \frac{1}{2} + \frac{\cos 4\theta}{2} \right] \cdot d\theta.$$

$$= \int_{\pi/4}^{\pi/2} \left[\frac{3}{2} + 2\cos 2\theta + \frac{\cos 4\theta}{2} \right] \cdot d\theta$$

$$= \left(\frac{3\theta}{2} + 2 \frac{\sin 2\theta}{2} + \frac{\sin 4\theta}{2 \times 4} \right) \Big|_{\pi/4}^{\pi/2}$$

$$= \left(\frac{3\theta}{2} + \sin 2\theta + \frac{\sin 4\theta}{8} \right) \Big|_{\pi/4}^{\pi/2}$$

$$= \left(\frac{3}{2} \cdot \frac{\pi}{2} + \sin 2 \times \frac{\pi}{2} + \frac{\sin 4 \times \frac{\pi}{2}}{8} - \left(\frac{3}{2} \cdot \frac{\pi}{4} + \sin 2 \times \frac{\pi}{4} + \frac{\sin 4 \times \frac{\pi}{4}}{8} \right) \right)$$

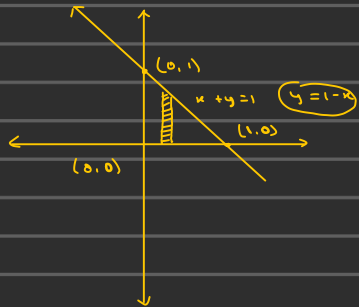
$$= 0 //$$

⑤ Limit is not given (Region)

Q.1) Find the integral $\iint_R (y - \sqrt{x}) \, dA$ where R is the region cut-off by the line $x + y = 1$ of the x - y plane in the 1st Quadrant.

$$\iint (y - \sqrt{x}) \, dA \, dy$$

u can take strip anyhow



$$x = 0$$

$$x = 1$$

$$y = 0$$

$$y = 1 - x$$

$$\int_0^1 \left[\int_0^{1-x} (y - \sqrt{x}) \, dy \right] dx$$

$$= \int_0^1 \left[\frac{y^2}{2} - \sqrt{x} \cdot y \right]_0^{1-x} dx$$

$$= \int_0^1 \left[\frac{(1-x)^2}{2} - \sqrt{x}(1-x) \right] dx$$

$$= \int_0^1 \left[\frac{1-2x+x^2}{2} - \sqrt{x} + \sqrt{x} \cdot x \right] dx$$

$$= \int_0^1 \left[\frac{1-2x+x^2 - 2\sqrt{x} + 2x^{3/2}}{2} \right] dx$$

$$= \frac{1}{2} \int_0^1 [1-2x+x^2 - 2\sqrt{x} + 2x^{3/2}] dx$$

$$= \frac{1}{2} \left[x - \frac{x^2}{2} + \frac{x^3}{3} - 2 \frac{x^{3/2}}{3/2} + \frac{2x^{5/2}}{5/2} \right]_0^1$$

$$= \frac{1}{2} \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{4}{3} x^{3/2} + \frac{4}{5} x^{5/2} \right]_0^1$$

$$= \frac{1}{2} \left[1 - 1 + \frac{1}{3} - \frac{4}{3} + \frac{4}{5} \right]$$

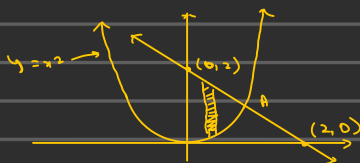
$$= \frac{1}{2} \left[\frac{1}{3} - \frac{4}{3} + \frac{4}{5} \right] = \frac{1}{2} \left[\frac{5 - 20 + 12}{15} \right]$$

$$= \frac{1}{2} \times \frac{-8}{15} = -\frac{4}{15}$$

Q1) Evaluate $\iint_R y \cdot dx \cdot dy$ over the area bounded by $x=0$, $y=x^2$, $x+y=2$ in the 1st Quadrant.

$$y=x^2$$

find A = (?)



$$x+y=2 \quad \& \quad y=x^2$$

$$x^2+x=2$$

$$x^2+x-2$$

$$(x+2)(x-1) = 0$$

$$(x+2)=0 \quad (x-1)=0$$

$$x = -2$$

$$x = 1$$

$$x=0$$

$$x=1$$

$$y=x^2$$

$$y=2-x$$

$$= \int_0^1 \left[\int_{x^2}^{2-x} y \cdot dy \right] \cdot dx$$

$$= \int_0^1 \left[\frac{y^2}{2} \right]_{x^2}^{2-x} dx$$

$$= \int_0^1 \left[\frac{(2-x)^2}{2} - \frac{(x^2)^2}{2} \right] \cdot dx$$

$$= \int_0^1 \left[\frac{4 - 4x + x^2 - x^4}{2} \right] \cdot dx$$

$$= \frac{1}{2} \int_0^1 [4 - 4x + x^2 - x^4] \cdot dx$$

$$= \frac{1}{2} \left[4x - \frac{4x^2}{2} + \frac{x^3}{3} - \frac{x^5}{5} \right]_0^1$$

$$= \frac{1}{2} \left[4x - 2x^2 + \frac{x^3}{3} - \frac{x^5}{5} \right]_0^1$$

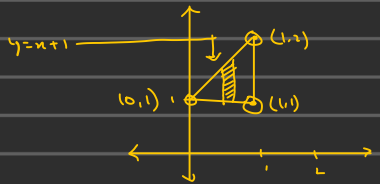
$$= \frac{1}{2} \left[4 - 2 + \frac{1}{3} - \frac{1}{5} \right]$$

$$= \frac{16}{15}$$

Q₂) Evaluate $\iint (x^2 + y^4) dx \cdot dy$ over the area of the Δ whose vertices are (0,1) (1,1) (1,2)

$$x=0 \quad x=1$$

$$y=1 \quad y=x+1$$



$$\int_0^1 \int_1^{x+1} (x^2 + y^2) dy \cdot dx$$

$$= \int_0^1 \left[x^2 \cdot y + \frac{y^3}{3} \right]_1^{x+1} dx$$

$$= \int_0^1 \left[x^2(x+1) + \frac{(x+1)^3}{3} - \left(x^2 + \frac{1}{3} \right) \right] dx$$

$(a+b)^3$
 $= a^3 + b^3 + 3a^2b + 3ab^2$

$$= \int_0^1 \left[\frac{x^3 + x^2 + 3x^2 + 3x + 1}{3} - x^2 - \frac{1}{3} \right] dx$$

$$= \int_0^1 \left[\frac{3x^3 + 3x^2 + x^2 + 3x + 1 - 3x^2 - 1}{3} \right] dx$$

$$= \frac{1}{3} \int_0^1 [4x^3 + 3x^2 + 3x] dx$$

$$= \frac{1}{3} \left[\frac{4 \cdot x^4}{4} + \frac{3x^3}{3} + \frac{3x^2}{2} \right]_0^1$$

$$= \frac{1}{3} \times \frac{7}{2} = \frac{7}{6}$$

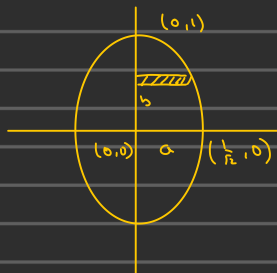
Q4) $\iint_R \frac{dx \cdot dy}{\sqrt{1-x^2-y^2}} = \frac{\pi}{4}$ where R is the region of the 1st Quadrant of the ellipse $x^2 + y^2 = 1$

$$2x^2 + y^2 = 1$$

$$\frac{x^2}{\frac{1}{2}} + \frac{y^2}{1} = 1$$

$$2x^2 = 1 - y^2$$

$$x = \sqrt{\frac{1-y^2}{2}}$$



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a^2 = \frac{1}{2} \quad b^2 = 1$$

$$a = \frac{1}{\sqrt{2}}$$

$$b = 1$$

$$x = 0$$

$$y = 0$$

$$x = \sqrt{1-y^2/2}$$

$$x = 1$$

$$= \int_0^1 \left[\int_0^{\sqrt{\frac{1-y^2}{2}}} \frac{1}{\sqrt{(1-y^2)^2 + x^2}} dx \right] dy$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right)$$

$$= \int_0^1 \left[\sin^{-1} \left(\frac{x}{\frac{1}{\sqrt{2}}}} \right) \right]_0^{\sqrt{\frac{1-y^2}{2}}} dy$$

$$= \int_0^1 \left[\sin^{-1} \left(\frac{\sqrt{1-y^2}}{\frac{1}{\sqrt{2}}}} \right) \right] dy$$

$$= \int_0^1 \left[\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) \right] dy = \int_0^1 \left[\frac{\pi}{4} \right] dy$$

$$= \left[\frac{\pi}{4} \cdot y \right]_0^1 = \frac{\pi}{4} \quad \text{Hence Proved.}$$

Q₂) Evaluate $\iint_R xy \cdot dxdy$ over the region R given by $x^2 + y^2 - 2x = 0$, $y^2 = 2x$
 $y = x$.

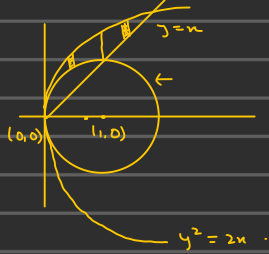
$$x^2 + y^2 - 2x = 0$$

$$x^2 - 2x + 1 + y^2 = 1$$

↓

$$(x-1)^2 + y^2 = 1^2$$

$$\text{centre} = (1, 0)$$



$$y^2 = 2x$$

$$x^2 + y^2 - 2x = 0$$

$$2x^2 - 2x = 0$$

$$x=0$$

$$x=1$$

$$2x(x-1) = 0$$

$$2x = 0 \quad x = 1$$

$$y^2 = 2x \quad \& \quad y = 4$$

$$x^2 = 2$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x=0$$

$$\& \quad x=2$$

$$y^2 = \sqrt{2x - x^2}$$

$$\int_0^1 \left[\int_{\sqrt{2x-x^2}}^{\sqrt{2x}} xy \cdot dy \cdot dx + \int_1^2 \left[\int_x^{\sqrt{2x}} xy \cdot dy \right] \cdot dx \right]$$

6

Q1) Calculate $\int (r^2 dr d\theta)$ over the area included b/w the circles
 $r = 2\sin\theta$ & $r = 4\sin\theta$.

$$r = 2\sin\theta$$

$$r^2 = 2r\sin\theta$$

$$x^2 + y^2 = 2y$$

$$x^2 + y^2 - 2y = 0$$

$$x^2 + y^2 - 2(y)(1) + 1^2 = 1^2$$

$$x^2 + (y-1)^2 = 1^2$$

$$C \equiv (0,1)$$

$$r = 4\sin\theta$$

$$r^2 = 4r\sin\theta$$

$$x^2 + y^2 = 4y$$

$$x^2 + y^2 - 4y = 0$$

$$x^2 + y^2 - 2(4)(2) + 2^2 = 2^2$$

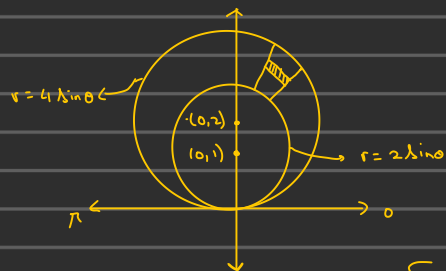
$$x^2 + (y-2)^2 = 2^2$$

$$C \equiv (0,2)$$

$$y = r\sin\theta$$

$$x = r\cos\theta$$

$$x^2 + y^2 = r^2$$



$$A = \int_0^\pi \left[\int_{r=2\sin\theta}^{r=4\sin\theta} r^2 \cdot dr \right] \cdot d\theta$$

$$= \int_0^\pi \left[\frac{r^4}{4} \right]_{2\sin\theta}^{4\sin\theta} d\theta = \frac{1}{4} \int_0^\pi [4^4 \sin^4\theta - 2^4 \sin^4\theta] d\theta$$

$$= \frac{1}{4} \times \pi \int_0^\pi [4^3 \sin^4\theta - 2^3 \sin^4\theta] d\theta$$

$$= \int_0^{\pi} [4^3 - 4] \sin^4 \theta \cdot d\theta$$

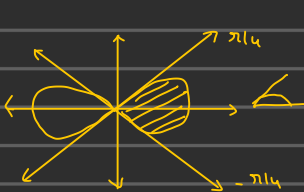
$$= 60 \int_0^{\pi} \sin^4 \theta \cdot d\theta$$

$$= 60 \times 2 \int_0^{\pi/2} \sin^4 \theta \cdot d\theta$$

$$= 120 \times \frac{3}{4} \cdot \frac{1}{2} \times \frac{\pi}{2}$$

$$= 45\pi$$

Q2) Evaluate $\iint \frac{r \cdot d\theta \cdot dr}{\sqrt{r^2 + a^2}}$ over one loop of the lemniscate $r^2 = a^2 \cos 2\theta$



$$A = 2 \times \int_0^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} \frac{r \cdot d\theta \cdot dr}{\sqrt{r^2 + a^2}}$$

$$\int_{-\pi/4}^{\pi/4} \int_0^{a\sqrt{\cos 2\theta}} \frac{r \cdot d\theta \cdot dr}{\sqrt{r^2 + a^2}}$$

$$= 2 \times \int_0^{\pi/4} \left[\int_0^{a\sqrt{\cos 2\theta}} \frac{2r}{\sqrt{r^2 + a^2}} dr \right] \cdot d\theta$$

$$\frac{f'(u)}{f(u)} = \frac{2}{\sqrt{1+u^2}}$$

$$= 2 \int_0^{\pi/4} \left[\frac{1}{2} \times 2 \sqrt{r^2 + a^2} \right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= 2 \int_0^{\pi/4} \left[\sqrt{r^2 + a^2} \right]_0^{a\sqrt{\cos 2\theta}} d\theta$$

$$= 2 \times \int_0^{\pi/4} \left[\sqrt{(\cos \theta)^2 + a^2} - \sqrt{a^2} \right] \cdot d\theta$$

$$= 2 \times \int_0^{\pi/4} \left[\sqrt{a^2 \cos^2 \theta + a^2} - a \right] \cdot d\theta$$

$$= 2 \times \int_0^{\pi/4} \left[\sqrt{a^2 (\cos^2 \theta + 1)} - a \right] \cdot d\theta$$

$$= 2 \times \int_0^{\pi/4} \left[a \sqrt{\cos^2 \theta + 1} - a \right] \cdot d\theta \quad | + \cos^2 \theta = 2 \cos^2 \theta$$

$$= 2a \int_0^{\pi/4} \left[\sqrt{\cos^2 \theta + 1} - 1 \right] \cdot d\theta$$

$$= 2a \int_0^{\pi/4} \left[\sqrt{2 \cos^2 \theta} - 1 \right] \cdot d\theta$$

$$= 2a \int_0^{\pi/4} \left[\sqrt{2} \cos \theta - 1 \right] \cdot d\theta$$

$$= 2a \left[\sqrt{2} \sin \theta - \theta \right]_0^{\pi/4}$$

$$= 2a \left[\sqrt{2} \sin \frac{\pi}{4} - \frac{\pi}{4} \right]$$

$$= 2a \left[\sqrt{2} \times \frac{1}{\sqrt{2}} - \frac{\pi}{4} \right] = 2a \left(1 - \frac{\pi}{4} \right)$$

⑦ Circle / Bernoulli Lemniscate

Q1) Evaluate $\iint y^2 \cdot dA$ over the area outside $x^2 + y^2 - a^2 = 0$ & inside $x^2 + y^2 - 2ax = 0$

$$x^2 + y^2 - a^2 = 0$$

$$x^2 - 2ax + y^2 = 0$$

$$x^2 - 2(x)\left(\frac{a}{2}\right) + y^2 = 0$$

$$x^2 - 2(x)\left(\frac{a}{2}\right) + \frac{a^2}{4} + y^2 = \frac{a^2}{4}$$

$$(x - a/2)^2 + y^2 = a^2/4$$

$$\text{Centre} = (a/2, 0)$$

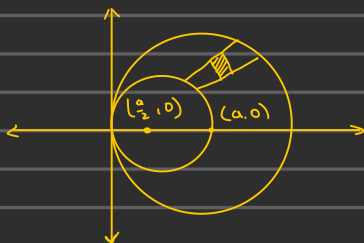
$$x^2 + y^2 - 2ax = 0$$

$$x^2 - 2ax + y^2 = 0$$

$$x^2 - 2(x)(a) + a^2 + y^2 = a^2$$

$$(x-a)^2 + y^2 = a^2$$

$$\text{Centre} = (a, 0)$$



$$x^2 + y^2 = a^2$$

$$r^2 = a^2 \cos^2 \theta$$

$$r = a \cos \theta$$

$$x^2 + y^2 = 2ax$$

$$r^2 = 2ax$$

$$r = 2a \cos \theta$$

$$A = 2 \times \int_0^{\pi/2} \int_{a \cos \theta}^{2a \cos \theta} r^2 \sin^2 \theta \cdot r \, dr \, d\theta$$

$$A = 2 \times \int_0^{\pi/2} \int_{a \cos \theta}^{2a \cos \theta} r^3 \sin^2 \theta \, dr \, d\theta$$

$$A = 2 \times \int_0^{\pi/2} \left[\frac{\sin^2 \theta \cdot r^4}{4} \right]_{a \cos \theta}^{2a \cos \theta} d\theta$$

$$A = \int_0^{\pi/4} \int \left[\sin^2 \theta - 2a^2 \cos^4 \theta - \sin^2 \theta \cdot a^4 \cos^4 \theta \right] \cdot d\theta$$

$$A = \frac{a^4}{2} \int_0^{\pi/4} \left[16 \sin^2 \theta \cdot \cos^4 \theta - \sin^2 \theta \cos^4 \theta \right] \cdot d\theta$$

$$A = \frac{a^4}{2} \int_0^{\pi/4} [16 - 1] \sin^2 \theta \cdot \cos^4 \theta d\theta$$

$$A = \frac{15}{2} a^4 \int_0^{\pi/4} \sin^2 \theta \cdot \cos^4 \theta \cdot d\theta$$

$$A = \frac{15a^4}{4} \cdot \frac{1 \times 3 \times 1}{6 \times 4 \times 2} \times \frac{\pi}{2}$$

$$A = \frac{15\pi a^4}{64}$$

Q1) change to polar coordinate & evaluate $\iint_R \frac{dx \cdot dy}{(1+x^2+y^2)^2}$ over one

loop of Lemniscate $(x^2+y^2)^2 = x^2 - y^2$

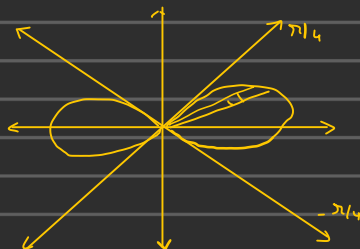
$$(x^2+y^2)^2 = x^2 - y^2$$

$$(r^2)^2 = r^2 \cos^2 \theta - r^2 \sin^2 \theta$$

$$r^2 = r^2 (\cos^2 \theta - \sin^2 \theta)$$

$$r^2 = \cos 2\theta$$

$$r = \sqrt{\cos 2\theta}$$



$$A = 2 \times \int_0^{\pi/4} \int_0^{\sqrt{\cos 2\theta}} \frac{r \cdot dr \cdot d\theta}{(1+r^2)^2}$$

$$A = 2 \int_0^{\pi/4} \left[\int_0^{\sqrt{\cos 2\theta}} \frac{r \cdot dr}{(1+r^2)^2} \right] \cdot d\theta$$

Put $1+r^2 = t$

$$2r \cdot dr = dt$$

$$r \cdot dr = \frac{1}{2} \cdot dt$$

$r=0$ $t=1$

$r = \sqrt{\cos 2\theta}$ $t = 1 + \cos 2\theta$

$$A = 2 \times \int_0^{\pi/4} \left[\int_1^{1+\cos 2\theta} \frac{1}{2} \cdot \frac{1}{t^2} \cdot dt \right] \cdot d\theta$$

$$A = \int_0^{\pi/4} \left[\frac{-1}{1+t} \right]_1^{1+\cos 2\theta} d\theta$$

$$\int \frac{1}{x^n} = \frac{-1}{(n-1)x^{n-1}}$$

$$A = - \int_0^{\pi/4} \left[\frac{1}{t} \right]_1^{1+\cos 2\theta} d\theta$$

$$A = - \int_0^{\pi/4} \left[\frac{1}{1+\cos 2\theta} - 1 \right] \cdot d\theta$$

$$1 + \cos 2\theta = 2 \cos^2 \theta$$

$$A = - \int_0^{\pi/4} \left[\frac{1}{2 \cos^2 \theta} - 1 \right] \cdot d\theta$$

$$A = - \int_0^{\pi/4} \left[\frac{1}{2} \sec^2 \theta - 1 \right] \cdot d\theta$$

$$A = \int_0^{\pi/4} \left(1 - \frac{1}{2} \sec^2 \theta \right) \cdot d\theta$$

$$A = \left[\theta - \frac{1}{2} \tan \theta \right]_0^{\pi/4}$$

$$A = \left[\frac{\pi}{4} - \frac{1}{2} (1) \right] = \frac{\pi}{4} - \frac{1}{2} = \boxed{\frac{\pi - 2}{4}}$$

⑧ Area by double Integrat ion

Q1) Find by double Integration the area bwn the curve $y^2 = 4x$ & $2x - 3y + 4 = 0$

$$y^2 = 4x$$

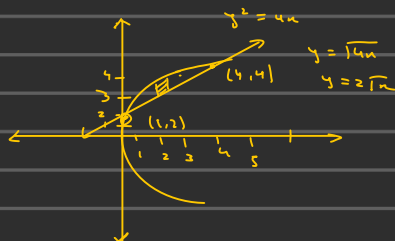
$$\left(\frac{2x+4}{3}\right)^2 = 4x$$

$$\frac{(2x+4)^2}{9} = 4x$$

$$2x - 3y + 4 = 0$$

$$2x + 4 = 3y$$

$$\boxed{\frac{2x+4}{3} = y}$$



$$(2x)^2 + 2(2x)(4) + 4^2 = 36x$$

$$4x^2 + 16x + 16 - 36x = 0$$

$$4x^2 - 20x + 16 = 0$$

$$x^2 - 5x + 4 = 0$$

$$(x-4)(x-1) = 0$$

$$\boxed{x = 4, 1}$$

$$x = 4 \quad y = \frac{2(4) + 4}{3} = 4$$

$$(4, 4)$$

$$x = 1 \quad y = \frac{2(1) + 4}{3} = \frac{6}{3} = 2$$

$$(1, 2)$$

$$A = \int_{x=1}^4 \int_{y=\frac{2x+4}{3}}^{y=\sqrt{4x}} dy \cdot dx$$

$$A = \int_1^4 \left[y \right]_{\frac{2x+4}{3}}^{\sqrt{4x}} dx$$

$$A = \int_1^4 \left[2\sqrt{x} - \left(\frac{2x+4}{3} \right) \right] \cdot dx$$

$$A = \int_1^4 \left[\frac{6\sqrt{x} - 2x - 4}{3} \right] \cdot dx$$

$$A = \frac{1}{3} \int_1^4 [6\sqrt{x} - 2x - 4] \cdot dx$$

$$A = \frac{1}{3} \left[6 \cdot \frac{2}{3} x^{3/2} - \frac{2 \cdot x^2}{2} - 4x \right]_1^4$$

$$A = \frac{1}{3} (4 \cdot 8 - 16 - 16) - (4 - 1 - 4)$$

$$A = \frac{1}{3} [(32 - 32) - (-1)]$$

$$A = \frac{1}{3} [0 + 1] = \frac{1}{3} //$$

Q.2) Find by double Integration the area btwn the parabola $y = x^2 - 6x + 3$ & line $y = 2x - 9$

$$y = x^2 - 6x + 3$$

$$y = x^2 - 2(x) + 3^2 + 3 - 3^2$$

$$y = (x-3)^2 + 3 - 9$$

$$y = (x-3)^2 - 6$$

$$y + 6 = (x-3)^2$$

$$(3, -6)$$

$$y = x^2 - 6x + 3$$

$$2x - 9 = x^2 - 6x + 3$$

$$= x^2 - 6x - 2x + 3 + 9$$

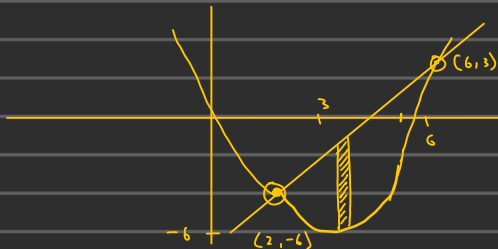
$$0 = x^2 - 8x + 12$$

$$0 = (x-6)(x-2)$$

$$x = 6, 2$$

$$x = 2, y = -5 \quad (2, -5)$$

$$x = 6, y = 3 \quad (6, 3)$$



$$A = \int_2^6 \int_{x^2-6x+3}^{2x-9} dy \cdot dx.$$

$$= \int_2^6 \left[y \right]_{x^2-6x+3}^{2x-9} dx$$

$$= \int_2^6 \left[(2x-9) - (x^2-6x+3) \right] \cdot dx$$

$$= \int_2^6 [2x-9-x^2+6x-3] dx$$

$$= \int_2^6 [8x-x^2-12] \cdot dx$$

$$A = \left[8 \times \frac{x^2}{2} - \frac{x^3}{3} - 12x \right]_2^6$$

$$A = \left[4x^2 - \frac{x^3}{3} - 12x \right]_2^6$$

$$A = \left\{ 4 \times 36 - \frac{216}{3} - 12 \times 6 \right\} - \left\{ 4(4) - \frac{8}{3} - 12(2) \right\}$$

$$A = \left[144 - \frac{216}{3} - 72 \right] - \left[16 - \frac{8}{3} - 24 \right]$$

$$A = \left[72 - \frac{216}{3} \right] - \left[-8 - \frac{8}{3} \right]$$

$$A = \left[7\frac{2}{3} - 7\frac{2}{3} + 8 + \frac{8}{3} \right] = \underline{\underline{\frac{32}{3}}}$$

Q3) Find by double Integration the area of the cardioid. $r = a(1 + \cos \theta)$

$$= 2 \times \int_0^{\pi} \int_0^{a(1+\cos \theta)} r \cdot dr \cdot d\theta$$

$$= 2 \int_0^{\pi} \left[\frac{r^2}{2} \right]_0^{a(1+\cos \theta)} d\theta$$

$$= 2 \int_0^{\pi} \left[\frac{a^2 (1+\cos \theta)^2}{2} \right] \cdot d\theta$$

$$= a^2 \int_0^{\pi} [1 + 2\cos \theta + \cos^2 \theta] \cdot d\theta$$

$$= a^2 \int_0^{\pi} \left[\left(2\cos^2 \frac{\theta}{2} \right)^2 \right] \cdot d\theta$$

$$= a^2 \int_0^{\pi} \left[\cos^4 \left(\frac{\theta}{2} \right) \right] \cdot d\theta$$

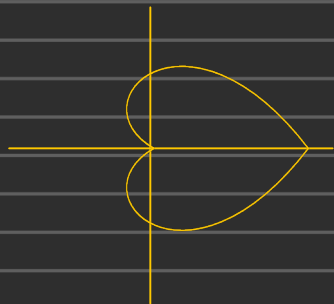
$$\text{Put } \theta/2 = t$$

$$\theta = 2t$$

$$\boxed{d\theta = 2dt}$$

$$\theta = 0 \quad t = 0$$

$$\theta = \pi \quad t = \pi/2$$



Triple Integrals

01

$$\int \frac{1}{u^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + C$$

$$\int \sec^2 u \cdot du = \tan u + C$$

$$\int \frac{1}{a^2 - u^2} du = \sin^{-1}\left(\frac{u}{a}\right) + C$$

$$\begin{matrix} x= & y= & z= \\ 1 & 3 & xy \\ \int & \int & \int \\ x=0 & y=0 & z=0 \end{matrix}$$

$dx \, dy \, dz \rightarrow$ limits will be x, y, z respectively.

Q.1) Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x \, dz \, dy \, dx$

$$\begin{matrix} y=1 & x=1 & z=1-x \\ \int & \int & \int \\ y=0 & x=y^2 & z=0 \end{matrix}$$

$$I = \int_{y=0}^1 \int_{x=y^2}^1 \int_{z=0}^{1-x} x \, dz \, dx \, dy$$

$$I = \int_0^1 \int_{y^2}^1 \left[\int_0^{1-x} x \, dz \right] dx \, dy$$

$$\int_0^1 \int_{y^2}^1 \left[x^2 \right]_0^{1-x} dx \cdot dy$$

$$I = \int_0^1 \int_{y^2}^1 x [(1-x) - 0] dx \cdot dy$$

$$I = \int_0^1 \int_{y^2}^1 x(1-x) dx \cdot dy$$

$$I = \int_0^1 \int_{y^2}^1 (x - x^2) dx \cdot dy$$

$$I = \int_0^1 \left[\int_{y^2}^1 (x - x^2) dx \right] dy$$

$$I = \int_0^1 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{y^2}^1 dy$$

$$I = \int_0^1 \left[\left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{y^4}{2} - \frac{y^6}{3} \right) \right] dy$$

$$I = \int_0^1 \left[\left(\frac{3 \cdot 2}{6} \right) - \left[\frac{y^4}{2} - \frac{y^6}{3} \right] \right] dy$$

$$I = \int_0^1 \left[\frac{1}{6} - \frac{y^4}{2} + \frac{y^6}{3} \right] dy$$

$$I = \left[\frac{1}{6} y - \frac{1}{2} \frac{y^5}{5} + \frac{1}{3} \frac{y^7}{7} \right]_0^1$$

$$I = \left[\frac{1}{6} - \frac{1}{60} + \frac{1}{21} \right]$$

Q) Evaluate $\int_0^1 \int_1^2 \int_0^{y^2} xyz \, dx \, dy \, dz$

$$I = \int_{z=0}^{z=1} \int_{y=1}^{y=2} \int_{x=0}^{x=y^2} xyz \, dx \, dy \, dz$$

$$I = \int_0^2 \int_1^2 \left[\int_0^{y^2} xyz \, dx \right] \cdot dy \cdot dz$$

$$I = \int_0^2 \int_1^2 \left[yz \frac{x^2}{2} \right]_0^{y^2} \cdot dy \cdot dz$$

$$I = \int_0^2 \int_1^2 \frac{y^2}{2} \left[(yz)^2 \right] dy \cdot dz$$

$$I = \frac{1}{2} \int_0^2 \int_{y^2=1}^{y^2=2} [y^3 z^2] dy \cdot dz$$

$$I = \frac{1}{2} \int_0^2 \left[\int_1^2 y^3 z^2 dy \right] \cdot dz$$

$$I = \frac{1}{2} \int_0^2 \left[z^2 \frac{y^4}{4} \right]_1^2 dz$$

$$J = \frac{1}{2} \int_0^2 z^3 [z^4 - 1] \cdot dz.$$

$$J = \frac{1}{8} \int_0^4 z^3 (16 - 1) dz$$

$$J = \frac{15}{8} \int_0^2 z^3 dz$$

$$J = \frac{15}{8} \left[\frac{z^4}{4} \right]_0^2 = \underline{\underline{\frac{15}{2}}}.$$

Lecture - 02

Q.)
$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dz \, dy \, dx}{\sqrt{1-x^2-y^2-z^2}}$$

$$I = \int_{x=0}^1 \int_{y=0}^{\sqrt{1-x^2}} \int_{z=0}^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{(1-x^2-y^2)-z^2}} dz \, dy \, dx$$

using

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right)$$

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \left(\frac{z}{\sqrt{1-x^2-y^2}} \right) \right]_0^{\sqrt{1-x^2-y^2}} dy \, dx$$

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \left(\frac{\sqrt{1-x^2-y^2}}{\sqrt{1-x^2-y^2}} \right) - \sin^{-1} \left(\frac{0}{\sqrt{1-x^2-y^2}} \right) \right] dy \, dx$$

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\frac{\pi}{2} \right] dy \, dx$$

$$I = \int_0^1 \left[\int_0^{\sqrt{1-x^2}} dy \right] dx$$

$$I = \int_0^1 \left[y \right]_0^{\sqrt{1-x^2}} dx$$

$$\int \sqrt{a^2-x^2} \, dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$I = \frac{\pi}{2} \int_0^1 \left(\sqrt{1-n^2} \right) dn$$

$$I = \frac{\pi}{2} \left[\frac{\pi}{2} \sqrt{1-n^2} + \frac{1}{2} \sin^{-1} \left(\frac{\pi}{1} \right) \right]_0^1$$

$$I = \frac{\pi}{2} \left[\frac{1}{2} \cdot \frac{\pi}{2} \right] = \boxed{\frac{\pi^2}{8}}$$

Q.2)

$$\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$$

$$I = \int_{x=0}^{\log 2} \int_{y=0}^x \int_{z=0}^{x+y} e^x \cdot e^y \cdot e^z dz dy dx$$

$$I = \int_0^{\log 2} \int_0^x \left[\int_0^{x+y} e^x \cdot e^y \cdot e^z dz \right] dy dx$$

$$I = \int_0^{\log 2} \int_0^x \left[e^x \cdot e^y \cdot e^z \right]_0^{x+y} dy dx$$

$$I = \int_0^{\log 2} \int_0^x \left[e^x \cdot e^y (e^{x+y} - e^0) \right] dy dx$$

$$I = \int_0^{\log 2} \int_0^x \left[e^x \cdot e^y (e^{x+y} - 1) \right] dy dx$$

$$I = \int_0^{\log_2 2} \int_0^1 [e^{2x} \cdot e^{2y} (e^x \cdot e^y - 1)] \cdot dx \cdot dy$$

$$I = \int_0^{\log_2 2} \int_0^1 [e^{2x} \cdot e^{2y} - e^x \cdot e^y] \cdot dx \cdot dy$$

$$I = \int_0^{\log_2 2} \left[e^{2x} \cdot \frac{e^{2y}}{2} - e^x \cdot e^y \right]_0^1 dx \cdot dy$$

$$I = \int_0^{\log_2 2} \left[\left(e^{2x} \cdot \frac{e^{2y}}{2} - e^x \cdot e^y \right) - \left(\frac{e^{2x}}{2} - e^x \right) \right] dx \cdot dy$$

$$I = \int_0^{\log_2 2} \left[\frac{e^{2x}}{2} - e^{2y} - \frac{e^{2x}}{2} + e^x \right] dx \cdot dy$$

$$I = \left[\frac{e^{2x}}{8} - \frac{e^{2x}}{2} - \frac{e^{2x}}{4} + e^x \right]_0^{\log_2 2} dy$$

$$I = \left[\left(\frac{e^{4 \log_2 2}}{8} - \frac{e^{2 \log_2 2}}{2} - \frac{e^{2 \log_2 2}}{4} + e^{\log_2 2} \right) - \left(\frac{1}{8} - \frac{1}{2} - \frac{1}{4} + 1 \right) \right] dy$$

$$I = \left[\left(\frac{e^{\log_2 2^4}}{8} - \frac{e^{\log_2 2^2}}{2} - \frac{e^{\log_2 2^2}}{4} + e^{\log_2 2} \right) - \left(\frac{1}{8} - \frac{1}{2} - \frac{1}{4} + 1 \right) \right] dy$$

$$I = \left(\frac{16}{8} - \frac{4}{2} \cdot \frac{4}{2} + 1 \right) - \left(\frac{1}{8} - \frac{1}{2} - \frac{1}{4} + 1 \right)$$

$$I = 5/8$$

③ Volume of tetrahedron

Q1) Evaluate $\iiint \frac{dx dy dz}{(1+x+y+z)^3}$ over the volume of tetrahedron
 $x=0, y=0, z=0, x+y+z=1$

$$\begin{aligned} x &= 0 & x &= 1 \\ y &= 0 & y &= 1-x \\ z &= 0 & z &= 1-x-y \end{aligned}$$

$$I = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(1+x+y+z)^3} dx dy dz$$

$$I = \int_0^1 \int_0^{1-x} \left[\int_0^{1-x-y} \frac{1}{(1+x+y+z)^3} dz \right] dx dy$$

$$I = \int_0^1 \int_0^{1-x} \left[\frac{-1}{(3-y)(1+x+y+z)^2} \right]_0^{1-x-y} dx dy$$

$$I = \int_0^1 \int_0^{1-x} \left[\frac{-1}{2(1+x+y+z)^2} \right]_0^{1-x-y} dx dy$$

$$I = -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[\frac{1}{(1+x+y+1-x-y)^2} - \frac{1}{(1+x+y)^2} \right] dx dy$$

$$I = -\frac{1}{2} \int_0^1 \int_0^{1-u} \left[\frac{1}{u} - \frac{1}{(1+u+y)^2} \right] dy du$$

$$I = -\frac{1}{2} \int_0^1 \left[\frac{1}{4u} - \frac{1}{(1+u+u)^2} \right]_{y=0}^{1-u} du$$

$$= -\frac{1}{2} \int_0^1 \left[\frac{1}{4u} + \frac{1}{(1+u+u)^2} \right]_{y=0}^{1-u} du$$

$$I = -\frac{1}{2} \int_0^1 \left[\left(\frac{1}{4} (1-u) + \frac{1}{(1+u+1-u)} \right) - \left(0 + \frac{1}{1+u} \right) \right] du$$

$$I = -\frac{1}{2} \int_0^1 \left[\frac{1}{4} - \frac{u}{4} + \frac{1}{2} - \frac{1}{1+u} \right] du$$

$$I = -\frac{1}{2} \int_0^1 \left[\frac{3}{4} - \frac{u}{4} - \frac{1}{1+u} \right] du$$

$$I = -\frac{1}{2} \left[\frac{3}{4}u - \frac{1}{4} \frac{u^2}{2} - \log|1+u| \right]_0^1$$

$$I = -\frac{1}{2} \left[\left(\frac{3}{4} - \frac{1}{8} \cdot \log 2 \right) - (0 - 0 - \log 1) \right]$$

$$I = -\frac{1}{2} \left[\frac{5}{8} - \log 2 \right]$$

$$I = \frac{1}{2} \left[\log 2 - \frac{5}{8} \right]$$

④ Sphere

sphere

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$x^2 + y^2 + z^2 = r^2$$

Whole sphere

$$dx \, dy \, dz = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

multiply starting $\times 1$

$$r = 0 \text{ to } a$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

Q1) Evaluate $\iiint_V xyz(x^2 + y^2 + z^2) \, dx \, dy \, dz$ over the first octant of the sphere

Sphere

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dx \, dy \, dz = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$r = 0 \text{ to } a$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

$$I = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^a \left(r^3 \sin^2 \theta \cdot \cos \theta \cdot \sin \phi \cdot \cos \phi (r^2) r^2 \sin \theta \, dr \, d\theta \, d\phi \right)$$

$$I = \int_0^{\pi/4} \int_0^{\pi/2} \int_0^a r^7 \sin^2 \theta \cos \theta \cdot \sin \phi \cos \phi \, dr \, d\theta \, d\phi$$

$$I = \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi \int_0^{\pi/2} \sin^2 \theta \cos \theta \, d\theta \int_0^a r^7 \, dr$$

$$I = \frac{1}{2} \times \frac{2 \times 1}{4 \times 2} \times \left[\frac{r^8}{8} \right]_0^a$$

$$I = \frac{1}{8 \times 8} [r^8]_0^a$$

$$I = \frac{1}{64} [a^8]$$

$$\boxed{I = \frac{a^8}{64}}$$

⑤ Epilipsoid

$$x = a r \sin \theta \cos \phi$$

$$y = b r \sin \theta \sin \phi$$

$$z = c r \cos \theta$$

$$dx dy dz = abc r^2 \sin \theta dr d\theta d\phi$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1^2$$

$$r = 0 \text{ to } r$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

Q1) $I = \iiint \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}} dx dy dz$ over the volume of ellipsoid.

$$x = a r \sin \theta \cos \phi$$

$$y = b r \sin \theta \sin \phi$$

$$z = c r \cos \theta$$

$$dx \cdot dy \cdot dz = abc r^2 \sin \theta dr d\theta d\phi$$

$$r = 0 \text{ to } 1$$

$$\theta = 0 \text{ to } \pi/2$$

$$\phi = 0 \text{ to } \pi/2$$

$$I = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \sqrt{1} abc r^2 \sin \theta dr d\theta d\phi$$

$$I = \frac{1}{2} \int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin \theta d\theta \int_0^1 r^2 dr$$

$$I = 8abc \left[\overline{\phi} \right]_0^{\pi/2} (-\cos \theta)_0^{\pi/2} \left[\frac{1^3}{3} \right]_0^1$$

$$I = 8abc \left(\frac{\pi}{2} \right) [\cos \frac{\pi}{2} - \cos 0] \frac{1}{3} [1]$$

$$I = 8abc \left(\frac{\pi}{2} \right) \left(\frac{1}{3} \right) (0 - 1) = -\frac{4\pi abc}{3}$$

$$= \frac{4}{3} \pi abc$$

⑥ Numerical Solution of O.D.

E of 1st order & 1st °

Euler's method :-

Q1) use Euler's method to find an approximate value of y when $x=1.3$ in 3 steps $h=0.1$ given $\frac{dy}{dx} = \frac{y-x}{\ln y}$ & $y(1) = 2$

$$y(x_0) = 2$$

$$x_0 = 1$$

$$y_0 = 2$$

how to find $h = \frac{x - x_0}{n}$
 $n \leftarrow$ no. of steps

$$\frac{dy}{dx} = \frac{y-x}{\ln y}$$

$$f(x, y) = \frac{y-x}{\ln y}$$

$$x_1 = 1.1 \quad y_1 = y_0 + h f(x_0, y_0)$$

$$y_1 = 2 + 0.1 f(1, 2)$$

$$y_1 = 2 + 0.1 \times 0.7071$$

$$y_1 = 2.0707$$

$$y_n = y_{n-1} + h f(x_{n-1}, y_{n-1})$$

$$f(1, 2) = \frac{2-1}{\ln 2} = \frac{1}{\ln 2} \Rightarrow 0.707$$

$$x_2 = 1.2 \quad y_2 = y_1 + h f(x_1, y_1)$$

$$y_2 = 2.0707 + 0.1 f(1.1, 2.0707)$$

$$y_2 = 2.1350$$

$$x_3 = 1.3 \quad y_3 = y_2 + h f(x_2, y_2)$$

$$= 2.1350 + 0.1 f(1.2, 2.1350)$$

$$\Rightarrow y_3 = 2.1934$$

$$x_4 = 1.4$$

$$y_4 = y_3 + h f(x_3, y_3)$$

$$y_4 = 2.1934 + 0.1 f(1.3, 2.1934)$$

$$y_4 = 2.2463$$

$$x_5 = 1.5$$

$$y_5 = y_4 + h f(x_4, y_4)$$

$$y_5 = 2.2463 + 0.1 f(1.4, 2.2463)$$

$$y_5 = 2.2940$$

numerical integration

Trapezoidal rule

$$I = \frac{h}{2} [x + 2R]$$

xtreme

Simpson's 1/3rd rule:-

$$I = \frac{h}{3} [x + 2E + 4Q]$$

Even positions.

odd position (y_1, y_2)

Simpson's 3/8th rule:-

$$I = \frac{3h}{8} [x + 2T + 3R]$$

Remainder

Multiple of 3

$$I = \int_0^1 \frac{x}{1+x} dx \quad y = \frac{x}{1+x}$$

Initial values

$$h = \frac{UL - LL}{6}$$

6 ← no. of interval (in this case 6)

$$x \quad -1$$

$$y$$

$$y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$$

$$x = y_0 + y_6$$

R = Remainder

$$\text{if } \sqrt{\sin^2 + \cos^2} \rightarrow \text{1 radian mode}$$

$$\log_{10} 5 = 0.699 \quad \text{base 10 natural.}$$

$$\log 5 \Rightarrow \ln 5$$

Q) Evaluate $\int_{-1}^1 \frac{1}{1+x^2} dx$ by using :
 i) trapezoidal
 ii) Simpson ($1/3$ rd)
 iii) Simpson ($3/8$ th)

x	-1	$-2/3$	$-1/3$	0	$1/3$	$2/3$	1
$y = \frac{1}{1+x^2}$	$1/2$	$9/13$	$9/10$	1	$9/10$	$9/13$	$1/2$
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

$$h = \frac{1 - (-1)}{6} = \frac{1}{3}$$

Trapezoidal rule

$$I = \frac{h}{2} [x + 2R]$$

$$I = \frac{1/3}{2} [(y_0 + y_6) + 2(y_1 + y_2 + y_4 + y_5)]$$

$$I = \frac{1}{6} \left[\left(\frac{1}{2} + \frac{1}{2} \right) + 2 \left(\frac{9}{13} + \frac{9}{10} + 1 + \frac{9}{10} + \frac{9}{13} \right) \right]$$

$$I = 1.5615$$

ii) Simpson's $1/2$ rd rule.

$$I = h/3 [x + 2E + 4O]$$

$$I = 1/3 \left[(y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right]$$

$$I = \frac{1}{9} \left[\left(\frac{1}{2} + \frac{1}{2} \right) + 2 \left(\frac{9}{13} + \frac{9}{10} \right) + 4 \left(\frac{9}{13} + 1 + \frac{9}{13} \right) \right]$$

$$I = 1.5704$$

iii) Simpson $(3/8$ th) rule.

$$I = \frac{3h}{8} [x + 2\tau + 3R]$$

$$= \frac{3h}{8} [(y_0 + y_6) + 2(y_3) + 3(y_1 + y_2 + y_4 + y_5)]$$

$$= \frac{3h}{8} \left[\left(\frac{1}{2} + \frac{1}{2} \right) + 2(1) + 3 \left(\frac{9}{13} + \frac{9}{10} + \frac{9}{10} + \frac{9}{13} \right) \right]$$

Range kutta of 4th order

$$K_1 = hf(x_0, y_0)$$

$$K_2 = hf(x_0 + h/2, y_0 + K_1/2)$$

$$K_3 = hf(x_0 + h/2, y_0 + K_2/2)$$

$$K_4 = hf(x_0 + h, y_0 + K_3)$$

$$K = \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

	x_0	x_1	x_2
x	0	0.1	0.2
y	1	✓	✓
	y_0	y_1	y_2

$$y_1 = y_0 + K$$

find y at
 $x=0.2$

$$x_0 = 0 \quad y_0 = 1 \quad h = 0.1$$

$$y(0) = 1$$

$$x_0 = 0 \quad y_0 = 1$$

$$y_2 = y_1 + K$$

$y(0) = 1$ find y at $x=0.1$

$$x_0 = 0, y_0 = 1$$

x
y

y at $x = 0.2$
 find $\frac{dy}{dx} = x^2 + y$ by Runge Kutta method of 4th order for
 $y(0) = 2$
 $h = 0.2$

	x_0	x_1	x_2
x	0	0.2	
y	2	✓	
	y_0	y_1	y_2

Given $y(0) = 2$

$x_0 = 0$ $y_0 = 2$ $h = 0.2$

$$f(x, y) = x^2 + y$$

$$k_1 = h f(x_0 + h/2, y_0 + k_1/2)$$

$$\begin{aligned} k_1 &= h f(x_0, y_0) \\ &= 0.2 f(0, 2) \\ &= 0.2 [0^2 + 2] \end{aligned}$$

$$k_2 = 0.2 f\left(0 + \frac{0.2}{2}, 2 + \frac{0.4}{2}\right)$$

$$k_2 = 0.2 f(0.1, 2.2)$$

$$k_1 = 0.4$$

$$k_2 = 0.2 [0.1^2 + 2.2]$$

$$k_2 = 0.4402$$

$$k_3 = h f(x_0 + h/2, y_0 + k_3/2)$$

$$= 0.2 f\left(0 + \frac{0.2}{2}, 2 + \frac{0.4402}{2}\right)$$

$$= 0.2 f(0.1, 2.2201)$$

$$= 0.2 [0.1^2 + 2.2201]$$

$$k_3 = 0.4442$$

$$k_4 = h f(x_0 + h, y_0 + k_4)$$

$$= 0.2 f(0 + 0.2, 2 + 0.4442)$$

$$= 0.2 f(0.2, 2.4442)$$

$$= 0.2 [0.2^2 + 2.4442]$$

$$k_4 = 0.4904$$

$$K = \frac{1}{6} [K_1 + 2K_2 + 2K_3 + K_4]$$

$$= \frac{1}{6} [0.4 + 2(0.4402) + 2(0.4401) + 0.4404]$$

$$K = 0.4432$$

$$y_1 = y_0 + K$$

$$= 2 + 0.4432$$

$$y_1 = 2.4432$$

Euler's modified formula

$$y_1 = y_0 + h f(x_0, y_0)$$

$$y_1' = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)]$$

$$y_1^{(2)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$y_1^{(3)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1^{(2)})]$$

$$y_1^{(4)} =$$

Q) Use modified Euler's method find $y(0.1)$ for $dy/dx = x - y^2$, $y(0) = 1$ with $h = 0.2$ correct upto 3 decimal.

$$y_1 = y_0 + h f(x_0, y_0) \quad y_1^{(4)} = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)]$$

Runge kutta of 4th order

initial value problem $\frac{dy}{dx} = f(x, y)$ where $y(x_0) = y_0$

$$k_1 = hf(x_n, y_n)$$

$$k_2 = hf(x_{n+1/2}, y_n + k_1/2)$$

$$k_3 = hf(x_{n+1/2}, y_n + k_2)$$